

Developmental conversion of thymocyte-attracting cells into self-antigen-displaying cells in embryonic thymus medulla epithelium

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Abstract

Thymus medulla epithelium establishes immune self-tolerance and comprises diverse cellular subsets. Functionally relevant medullary thymic epithelial cells (mTECs) include a self-antigen-displaying subset that exhibits genome-wide promiscuous gene expression promoted by the nuclear protein Aire and that resembles a mosaic of extrathymic cells including mucosal tuft cells. An additional mTEC subset produces the chemokine CCL21, thereby attracting positively selected thymocytes from the cortex to the medulla. Both self-antigen-displaying and thymocyte-attracting mTEC subsets are essential for self-tolerance. Here we identify a developmental pathway by which mTECs gain their diversity in functionally distinct subsets. We show that CCL21-expressing mTECs arise early during thymus ontogeny. Fate-mapping analysis reveals that self-antigen-displaying mTECs, including Aire-expressing mTECs and thymic tuft cells, are derived from CCL21-expressing cells. The differentiation capability of CCL21-expressing embryonic mTECs is verified in reaggregate thymus experiments. These results indicate that CCL21-expressing embryonic mTECs carry a developmental potential to give rise to self-antigen-displaying mTECs, revealing that the sequential conversion of thymocyte-attracting subset into self-antigen-displaying subset serves to assemble functional diversity in the thymus medulla epithelium.

eLife assessment

This **important** study provides new insights into the development and function of medullary thymus epithelial cells (mTEC). The authors provide **compelling** evidence to support their claims as to the differentiation and lineage outcomes of CCL21+ mTEC progenitors, which further our understanding of how central tolerance of T cells is enforced within the thymus.

Introduction

The thymus medulla epithelium provides a unique environment for newly generated T cells to acquire self-tolerance before their emigration to the circulation. A variety of medullary thymic epithelial cells (mTECs) contribute to displaying self-antigens by employing multiple mechanisms. Promiscuous gene expression by a subpopulation of mTECs that express the nuclear protein Aire allows chromatin remodeling and genome-wide ectopic gene transcription^{1,2}. Cellular mimesis in other mTEC subpopulations relies on a mosaic of tissue-specific transcription factor-dependent cells that individually resemble specialized cell types, including mucosal tuft cells and microfold cells^{3,4}. These mTEC subpopulations form a self-antigen-displaying mTEC subset that provides various self-components, including extrathymic tissue-specific molecules, in the thymus medulla, so that newly generated T cells have an opportunity to establish central tolerance to genome-wide and systemic self-antigens prior to export from the thymus.

In addition to the self-antigen-displaying mTEC subset, there exists a medullary epithelial subset that produces the C-C chemokine CCL21, which attracts developing thymocytes and dendritic cells to the medullary region⁵. Positive selection-inducing TCR signals in newly generated cortical thymocytes elevate the expression of CCR7, a receptor for CCL21, and positively selected cortical thymocytes are therefore specifically attracted to migrate to the medullary region through the CCL21-CCR7-mediated chemotactic signals^{6,7}. Two molecular species CCL21Ser and CCL21Leu with one amino acid difference are encoded in the mouse genome^{8,9}, and CCL21Ser encoded by *Ccl21a* locus is predominantly expressed in the thymus medulla¹⁰. In mice lacking either *Ccr7* or *Ccl21a*, positively selected mature thymocytes fail to accumulate in the medulla, and therefore T cells fail to establish self-tolerance^{10–12}. Additional CCR7-ligand CCL19 has no appreciable role in the cortex-to-medulla migration of developing thymocytes^{10,13}. Thus, CCL21-expressing thymocyte-attracting mTECs represent a functional mTEC subset that is essential for the establishment of self-tolerance in T cells.

Aire-expressing mTECs primarily belong to MHC-II^{high} CD80^{high} mTEC subpopulation (mTEC^{high})^{14,15}, whereas most thymic mimetic cells are detectable in MHC-II^{low} CD80^{low} mTEC subpopulation (mTEC^{low}), which are derived from Aire-expressing mTEC^{high} cells^{4,16–18}. The mTEC^{low} subpopulation is heterogeneous and contains immature mTEC progenitors that differentiate into the Aire-expressing mTEC^{high} subpopulation^{19,20}. Consequently, the developmental progression of the self-antigen-displaying mTEC subset is considered to occur in the following sequence: mTEC^{low} progenitors → mTEC^{high} Aire-expressing cells → mTEC^{low} mimetic cells.

CCL21-expressing mTECs are mostly included in the mTEC^{low} subpopulation²¹. However, how the CCL21-expressing mTEC subset is integrated with, and/or potentially diverted from, the developmental progression of the self-antigen-displaying mTEC subset remains unknown. A single cell RNA-sequencing-based cluster analysis reported that *Ccl21a* transcripts are detectable within a heterogeneous “intertypical” TEC subpopulation, which includes cells with gene expression

profiles resembling immature mTEC progenitors²². A more recent study has indicated that postnatally appearing mTEC-biased progenitors include cells that transcribe *Ccl21a*²³. However, an independent single cell RNA-sequencing-based study suggested that TECs expressing *Ccl21a* transcripts represent a terminal population derived from “transit-amplifying” mTEC progenitors, by branching out of the self-antigen-displaying mTEC lineage²⁴. Without the direct evaluation of developmental potential, it remains controversial and unknown how CCL21-expressing mTECs are developmentally related to immature mTEC progenitors. It is even unclear how the *Ccl21a*-transcript-detectable TECs reported in these single cell transcriptomic studies are related to functionally relevant CCL21-protein-producing medullary TECs. More specifically, it remains unestablished whether the functionally relevant CCL21-expressing mTEC subset retains a developmental potential to give rise to the self-antigen-displaying mTEC subset.

The present study addresses whether the CCL21-expressing mTEC subset carries a developmental capability to differentiate into the self-antigen-displaying mTEC subset. We show that CCL21-expressing mTECs arise early during embryonic development in mice. By Cre-loxP-mediated fate-mapping analysis, we demonstrate that a vast majority of self-antigen-displaying mTEC subset, including Aire-expressing mTECs and thymic tuft cells, originates from CCL21-expressing cells. The developmental potential of CCL21-expressing embryonic mTECs to give rise to Aire-expressing mTECs is confirmed in reaggregate thymus experiments. These results indicate that functionally relevant CCL21-expressing mTEC subset contains a developmental capability to differentiate into self-antigen-displaying mTEC subset and that developmental conversion from thymocyte-attracting subset to self-antigen-displaying subset serves to assemble functional diversity in thymus medulla epithelium.

Results

CCL21-expressing mTECs arise in embryonic thymus

We began this study by examining the appearance of CCL21-expressing mTECs during mouse ontogeny. To do so, we analyzed CCL21-expressing mTECs by detecting tandem dimer Tomato (tdTomato) fluorescence proteins in *Ccl21a*-tdTomato knock-in mice. In the postnatal thymus of 5-week-old *Ccl21a*^{tdTomato/+} mice, CCL21-expressing cells were detected in keratin 14 (Krt14)-expressing medullary TECs (**Fig. 1A**) and not in Ly51-expressing cortical TECs (**Fig. 1B**). Flow cytometric analysis confirmed that *Ccl21a*^{tdTomato}⁺ cells in the thymus are predominantly confined in EpCAM⁺ Ly51[−] UEA1⁺ mTECs (**Fig. 1C, D**). In accordance with the medullary localization of CCL21-expressing mTECs, thymic medulla was abundant in CCL21 proteins (**Fig. 1E**), which contribute to attracting CCR7-expressing positively selected thymocytes from the cortex^{7,10}. Among mTECs, most CCL21-expressing mTECs were distinct from Aire-expressing mTECs and DCLK1⁺ thymic tuft cells (**Fig. 1F**). CCL21-expressing mTECs equally distributed throughout the medullary region and were enriched neither in the pericortical area, including the corticomedullary junction area, nor in the central area within the medulla (**Fig. 1G, H**, **Fig. S1A**).

Analysis of embryonic thymus showed that CCL21-expressing cells in the central region of the thymus were prominent by embryonic day 15 (E15) (**Fig. 2A**) and detectable as early as E13 (**Fig. 2B**). The localization of CCL21-expressing cells in the central region of embryonic thymus lobes and their expression of Krt14 and Foxn1 ascertained that CCL21-expressing embryonic thymus cells represented mTECs (**Fig. 2A–C**), rather than neighboring CCL21⁺ parathyroid gland cells (**Fig. 2B**), which contribute to the attraction of hematopoietic cells to the pre-vascularized embryonic thymus²⁵. CCL21 proteins were detectable in the medullary region proximal to *Ccl21a*^{tdTomato}-expressing mTECs in the embryonic thymus by E15 (**Fig. 2D**), indicating that *Ccl21a*^{tdTomato}-expressing E15 embryonic mTECs represent functionally relevant CCL21-expressing mTECs. On the other hand, CCL21-expressing mTECs in the embryonic thymus were largely distinct from Aire-expressing mTECs and DCLK1⁺ thymic tuft cells (**Fig. 2E**). Similar to

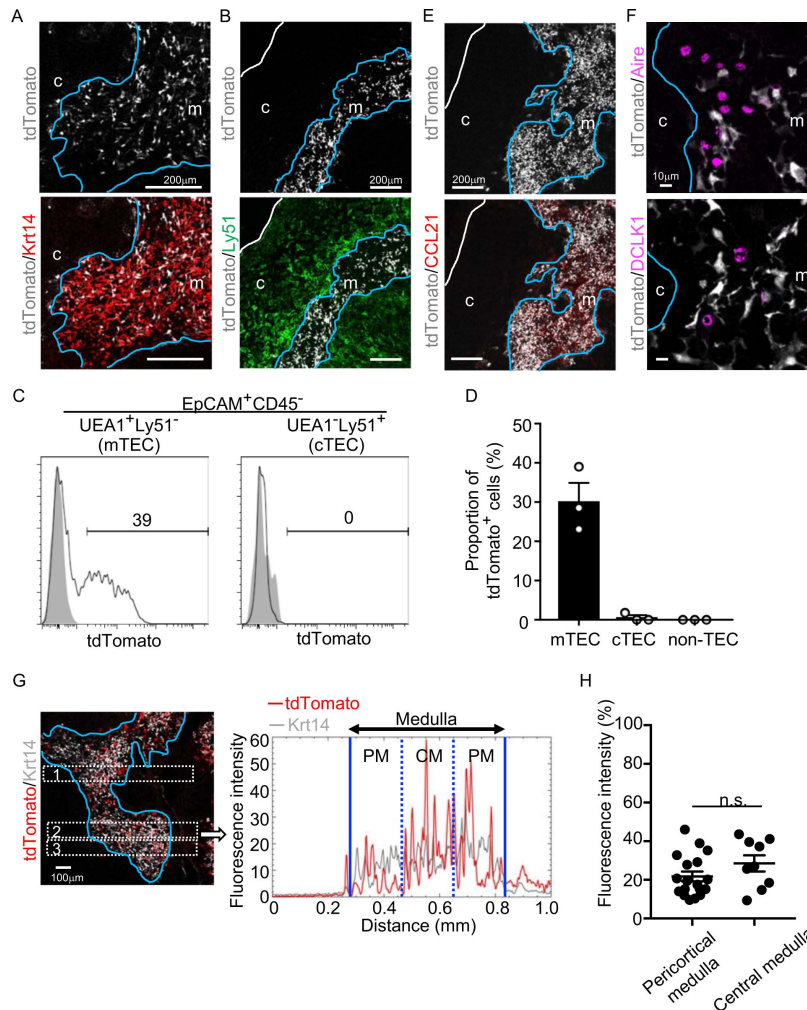


Figure 1.

CCL21-expressing mTECs in postnatal thymus

(A, B) Immunofluorescence analysis of thymus sections from 5-week-old *Ccl21a*^{tdTomato/+} mice. tdTomato fluorescence (white) was detected with Keratin 14 (Krt 14, red) (A) or Ly51 (green) (B). White lines indicate capsular outline of the thymus. Blue lines show cortico-medullary junction. Representative data from at least three independent experiments are shown. c, cortex. m, medulla. Scale bar, 200µm.

(C) Flow cytometric profiles of tdTomato fluorescence expression in EpCAM⁺CD45⁻ UEA1⁺Ly51⁻ mTECs (left) and EpCAM⁺CD45⁻ UEA1⁺Ly51⁺ cTECs (right) from *Ccl21a*^{tdTomato/+} mice (black line histograms) and WT mice (gray shaded histograms). Numbers in histograms indicate frequency of cells within indicated area.

(D) Proportions (means and SEMs, n=3) of tdTomato⁺ cells in EpCAM⁺CD45⁻ UEA1⁺Ly51⁻ mTECs, EpCAM⁺CD45⁻ UEA1⁺Ly51⁺ cTECs, and EpCAM⁺ non-TECs analyzed as in C.

(E, F) Immunofluorescence analysis of thymus sections from 5-week-old *Ccl21a*^{tdTomato/+} mice. tdTomato fluorescence (white) was detected with CCL21 (red) (E), Aire or DCLK1 (magenta) (F). White lines indicate capsular outline of the thymus. Blue lines show cortico-medullary junction. Representative data from at least three independent experiments are shown. c, cortex. m, medulla. Scale bar, 200µm (E) or 10µm (F).

(G) Distribution of tdTomato-expressing cells in the medullary region defined by Krt14 expression. Left panel shows representative tdTomato (red) and Krt14 (white) fluorescence signals in the thymus sections from *Ccl21a*^{tdTomato/+} mice. Scale bar, 100µm. Right panel shows fluorescence intensity profiles of tdTomato (red line) and Krt14 (gray line) signals within the region of interest defined by dashed rectangles in the left panel. Medullary regions were equally divided into three areas into pericortical areas (PM) and central area (CM).

(H) tdTomato intensity (means and SEMs, n=3) in indicated areas (G and Fig. S1A) were calculated in comparison with total tdTomato intensity within the ROI. n.s., not significant.

the postnatal thymus, CCL21-expressing mTECs in the embryonic thymus distributed throughout the medullary region and were not enriched in the pericortical areas including the corticomedullary junctions (**Fig. 2F, G** [Fig. S1B](#)). These results indicate that CCL21-expressing mTECs arise by E15 in the central region of the embryonic thymus.

Fate mapping of CCL21-expressing cells in thymus

To better understand whether CCL21-expressing mTECs have a developmental potential to give rise to the self-antigen-displaying mTEC subset, we engineered mice in which the P1 bacteriophage-derived *Cre* recombinase gene was knocked-in at *Ccl21a* locus (**Fig. 3A** [Fig. S1A](#)). We bred *Ccl21a*^{Cre/+} mice with mice that carried a ubiquitous CAG promoter-driven loxP-stop-loxP-EGFP (CAG-loxP-EGFP) transgene. In *Ccl21a*-Cre x CAG-loxP-EGFP mice, EGFP expression labeled cells that have transcribed *Ccl21a* previously and/or presently (**Fig. 3A** [Fig. S1A](#)).

In the postnatal thymus of 3-week-old *Ccl21a*-Cre x CAG-loxP-EGFP mice, EGFP signals in the thymus distributed prominently in the medullary region (**Fig. 3B** [Fig. S1B](#)). EGFP⁺ cells in the medulla were detected in Krt14⁺ mTECs (**Fig. 3B** [Fig. S1B](#)), including Aire⁺ mTECs and DCLK1⁺ thymic tuft cells (**Fig. 3C** [Fig. S1C](#)). Flow cytometric analysis of enzyme-digested thymus cells showed that EGFP⁺ cells in the thymus were detected exclusively in EpCAM⁺CD45⁻ TECs and not in EpCAM⁺CD45⁺ hematopoietic cells, including thymocytes and dendritic cells, or EpCAM⁺CD45⁻ cells, including mesenchymal fibroblasts and endothelial cells (**Fig. 3D, E** [Fig. S1D, S1E](#)). Within EpCAM⁺CD45⁻ TECs, approximately 95% of UEA1⁺Ly51⁻ mTECs were EGFP⁺, indicating that a vast majority of mTECs were derived from cells that previously transcribed *Ccl21a* (**Fig. 3F, G** [Fig. S1F, S1G](#)). Flow cytometric analysis also confirmed the detection of Aire⁺ UEA1⁺Ly51⁻ cells and DCLK1⁺ UEA1⁺Ly51⁻ cells in EpCAM⁺CD45⁻ EGFP⁺ TECs (**Fig. 3H** [Fig. S1H](#)). Thus, self-antigen-displaying mTECs, including Aire⁺ mTECs and DCLK1⁺ thymic tuft cells, which do not presently express CCL21 (**Fig. 1F** [Fig. S1I](#), 2E), originate from *Ccl21a*⁺ progenitors.

In the quantitative RT-PCR analysis of *Cre*, *Ccl21a*, and other TEC-associated genes, such as *Aire*, *Tnfrsf11a*, and *Psmb11*, *Cre* was detected specifically in the *Ccl21a*-expressing mTEC^{low} subpopulation, which includes the majority of CCL21-expressing mTECs²¹ [Fig. S1J](#), and not in the *Psmb11*-expressing cTECs or the *Aire*- and *Tnfrsf11a*-expressing mTEC^{high} subpopulation (**Fig. 4A** [Fig. S1K](#)). These results verify the lineage tracing potential of *Ccl21a*-Cre-expressing cells to differentiate into the mTEC^{high} subpopulation including Aire⁺ mTECs.

In addition to mTECs, approximately 66% of UEA1⁻Ly51⁺ cTECs in the postnatal thymus were EGFP⁺ in 3-week-old *Ccl21a*-Cre x CAG-loxP-EGFP mice (**Fig. 3F, G** [Fig. S1F, S1G](#)). Section analysis of the thymus verified the presence of EGFP⁺ cells in the cortex (**Fig. 3B** [Fig. S1B](#)). Because cTECs do not presently express *Ccl21a* transcripts¹⁰ [Fig. S1L](#), these results indicate that approximately two-thirds of postnatal cTECs are derived from cells that previously transcribed *Ccl21a* (**Fig. 3F, G** [Fig. S1F, S1G](#)). To characterize cTECs that previously transcribed *Ccl21a* and those that did not, we isolated EGFP⁺ and EGFP⁻ cTECs from 2-week-old postnatal *Ccl21a*-Cre x CAG-loxP-EGFP mice (**Fig. S2**) and performed deep-sequencing transcriptomic analysis. The MA plot and the volcano plot of RNA-sequencing data demonstrated high similarity in gene expression profiles between EGFP⁺ and EGFP⁻ cTECs among 20,230 genes detected (**Fig. 4B, C** [Fig. S1M, S1N](#)), unlike the pronounced difference in global gene expression profiles between cTECs and mTECs (**Fig. S3A, B**). The genes that are relevant for T cell-inducing and positive selection-inducing functions in cTECs, including *Dll4*, *Il7*, *Cxcl12*, *Psmb11*, *Ctsl*, *Prss16*, *Cd83*, *H-2K*, *H-2D*, and *I- β* , were similarly abundant in EGFP⁺ and EGFP⁻ cTECs (**Fig. S3C**). These results suggest that cTECs derived from cells that either do or do not transcribe *Ccl21a* are functionally comparable to promote the generation and positive selection of cortical thymocytes.

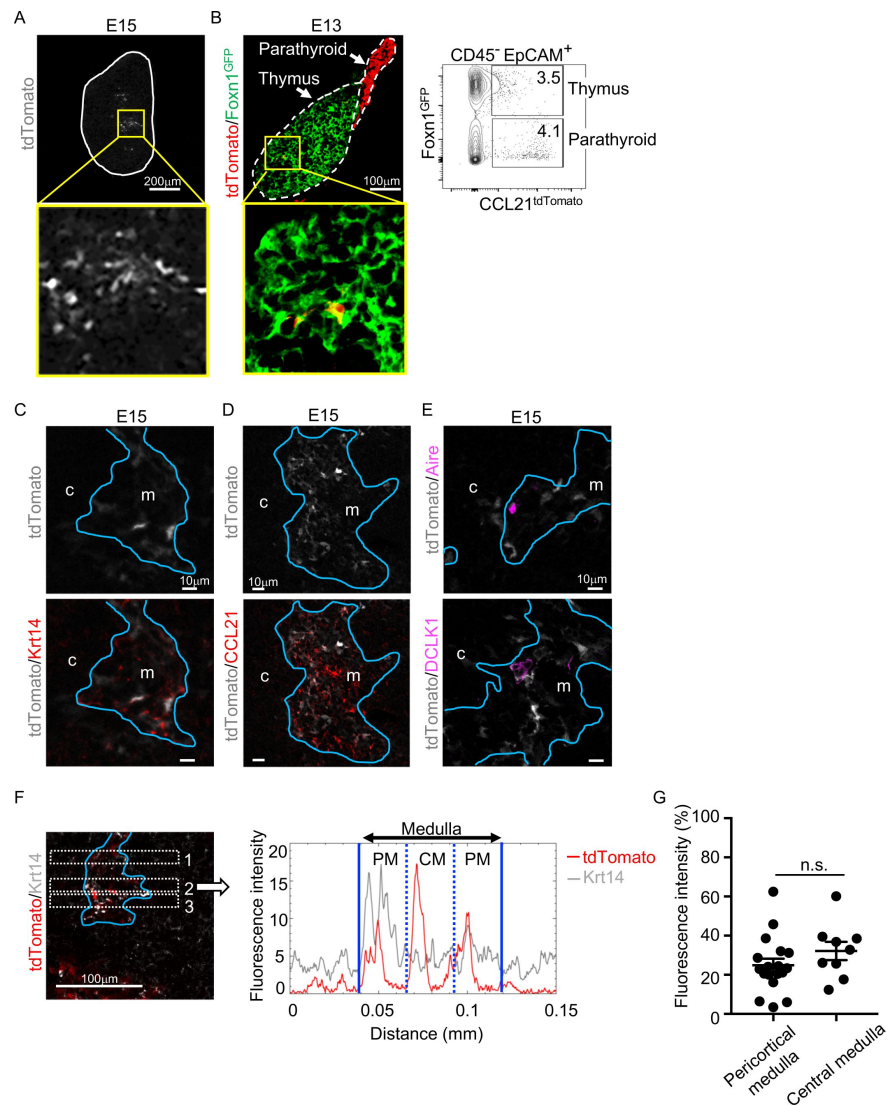


Figure 2.

CCL21-expressing mTECs in embryonic thymus

(A) tdTomato fluorescence (white) detected in thymus sections from E15 *Ccl21a^{tdTomato/+}* embryos. White lines indicate the capsular outline of the thymus. The image in central region identified by the box in upper panel is magnified in bottom panel. Representative data from at least three independent experiments are shown. Scale bar, 200µm.

(B) tdTomato (red) and GFP (green) fluorescence in thymus sections from E13 *Ccl21a^{tdTomato/+}* x Foxn1-GFP-transgenic embryos (left). Dashed lines show the outlines for the thymus and parathyroid primordia. Central region of the thymus identified by the box in upper panel is magnified in bottom panel. Scale bar, 100µm. Right panel shows a representative flow cytometric profile for tdTomato and GFP in CD45⁺EpCAM⁺ epithelial cells. Numbers in the contour plot indicate frequency of cells within indicated area.

(C-E) Immunofluorescence analysis of thymus sections from E15 *Ccl21a^{tdTomato/+}* embryos. tdTomato fluorescence (white) was detected with Krt14 (red) (C), CCL21 (red) (D), Aire, or DCLK1 (magenta) (E). Blue lines show the cortico-medullary junction. Representative data from at least three independent experiments are shown. c, cortex. m, medulla. Scale bar, 10µm.

(F) Distribution of tdTomato-expressing cells in the medullary region defined by Krt14 expression. Left panel shows representative tdTomato (red) and Krt14 (white) fluorescence signals in the thymus sections from *Ccl21a^{tdTomato/+}* E15 embryos. Scale bar, 100µm. Right panel shows fluorescence intensity profiles of tdTomato (red line) and Krt14 (gray line) signals within the region of interest defined by dashed rectangles in the left panel. The medullary regions were equally divided into three areas into pericortical areas (PM) and central area (CM).

(G) tdTomato intensity (means and SEMs, n=3) in indicated areas (F and Fig. S1B) were calculated in comparison with total tdTomato intensity within the ROI. n.s., not significant.

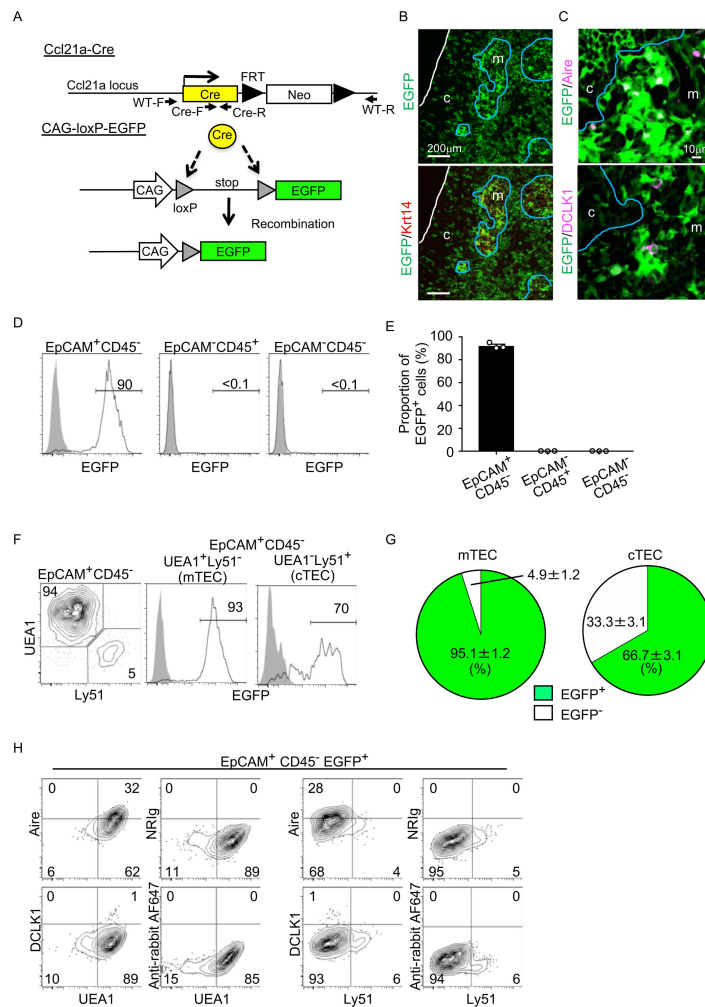


Figure 3.

Fate mapping of *Ccl21a*-expressing cells in the thymus

(A) Schematic illustration of Cre-dependent *Ccl21a*-specific EGFP expression in *Ccl21a*-Cre x CAG-loxP-stop-loxP-EGFP mice.

FRT, flippase recognition target site. Neo, neomycin resistance gene. Arrows indicate PCR primers for allele genotyping.

(B, C) Immunofluorescence analysis of thymus sections from 3-week-old *Ccl21a*-Cre x CAG-loxP-EGFP mice. EGFP fluorescence (green) was detected with Krt14 (red) (**B**), Aire, or DCLK1 (magenta) (**C**). White lines indicate the capsular outline of the thymus. Blue lines show the cortico-medullary junction. Representative data from at least three independent experiments are shown. c, cortex. m, medulla. Scale bar, 200µm (**B**) or 10µm (**C**).

(D) Flow cytometric analysis of liberase-digested thymus cells isolated from 3-week-old *Ccl21a*-Cre x CAG-loxP-EGFP mice. Histograms show EGFP fluorescence expression in EpCAM⁺CD45⁻ cells (left), EpCAM⁻CD45⁺ cells (middle), and EpCAM⁺CD45⁺ cells (right) prepared from *Ccl21a*-Cre x CAG-loxP-EGFP mice (black histograms) and littermate control (gray shaded histograms). Numbers in histograms indicate frequency of cells within indicated area.

(E) Proportions (means and SEMs, n=3) of EGFP⁺ cells in indicated cell populations analyzed as in **D**.

(F) Flow cytometric profiles of UEA1 reactivity and Ly51 expression in EpCAM⁺CD45⁻ TECs (left). Histograms show EGFP fluorescence expression in EpCAM⁺CD45⁻ UEA1⁺Ly51⁻ mTECs (middle) and EpCAM⁺CD45⁻ UEA1⁺Ly51⁺ cTECs (right) from *Ccl21a*-Cre x CAG-loxP-EGFP mice (black histograms) and littermate control (gray shaded histograms). Numbers in the histograms indicate frequency of cells within indicated area.

(G) Proportions (means and SEMs, n=3) of EGFP⁺ and EGFP⁻ cells in EpCAM⁺CD45⁻ UEA1⁺Ly51⁻ mTECs (left) and EpCAM⁺CD45⁻ UEA1⁺Ly51⁺ cTECs (right) analyzed as in **F**.

(H) Flow cytometric analysis of liberase-digested thymus cells isolated from *Ccl21a*-Cre x CAG-loxP-EGFP mice at 3 weeks old. Contour plots show Aire (top), DCLK1 (bottom), or isotype control detection with UEA1 reactivity or Ly51 expression in EpCAM⁺CD45⁻ EGFP⁺ TECs. Numbers in the plots indicate frequency of cells within the indicated area. Representative data from three independent experiments are shown.

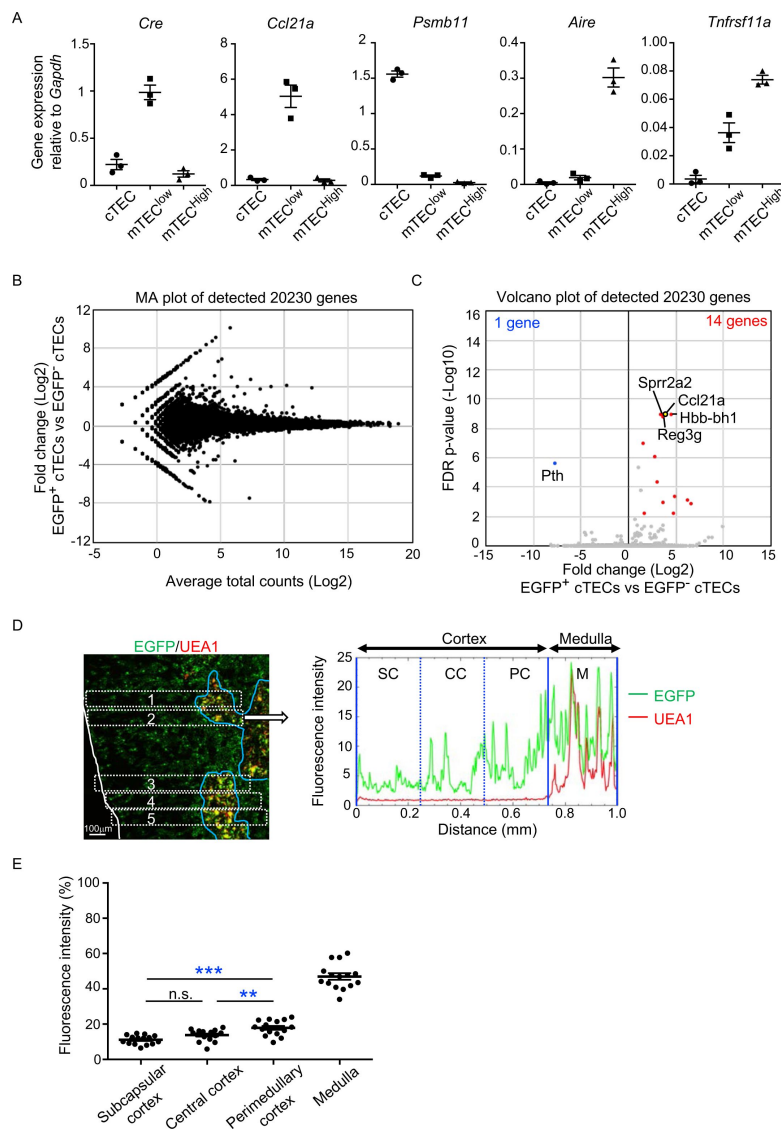


Figure 4.

Characterization of cTECs that previously transcribed *Ccl21a*

(A) Quantitative RT-PCR analysis of *Cre* and indicated genes (means and SEMs, n=3) in cTECs (EpCAM⁺CD45⁺UEA1⁻Ly51⁺), mTEC^{low} (EpCAM⁺CD45⁺UEA1⁺Ly51⁻I-A^{low}), and mTEC^{high} (EpCAM⁺CD45⁺UEA1⁺Ly51⁻I-A^{high}) isolated from *Ccl21a*^{Cre/+} mice.

(B) MA plot of 20,230 genes detected in RNA-sequencing analysis of EGFP⁺ and EGFP⁻ cTECs isolated from *Ccl21a*-Cre x CAG-loxP-EGFP mice at 2-week-old. Detected genes are plotted as log2 average total counts versus log2 fold changes (EGFP⁺ cTECs/EGFP⁻ cTECs).

(C) Volcano plot analysis of EGFP⁺ and EGFP⁻ cTECs. Detected genes are plotted as log2 fold changes (EGFP⁺ cTECs/EGFP⁻ cTECs) versus -log10 FDR p-values. Fourteen genes (red symbols) are more highly detected (log2 fold change > 1.7, FDR p-value < 0.05) in EGFP⁺ cTECs than EGFP⁻ cTECs, whereas one gene (blue symbol) is more highly detected (log2 fold change < -1.7, FDR p-value < 0.05) in EGFP⁻ cTECs than EGFP⁺ cTECs.

(D) Distribution of EGFP-expressing cells in the thymus. Left panel shows representative fluorescence signals for EGFP (green) in the thymus sections from *Ccl21a*-Cre x CAG-loxP-EGFP mice. The medullary region is defined by UEA1 reactivity (red). White lines indicate the capsular outline of the thymus. Blue lines show the cortico-medullary junction. Scale bar, 100µm. Right panel shows fluorescence intensity profiles of EGFP (green line) and UEA1 (gray line) signals within the region of interest (ROI) defined by dashed rectangles in the left panel. Cortical regions were equally divided into three areas into subcapsular cortex (SC), central cortex (CC), and perimedullary cortex (PC). M, medulla.

(E) EGFP intensity (means and SEMs, n=3) in indicated areas (**C** and **Fig. S1C**) were calculated in comparison with total EGFP intensity within the ROI. n.s., not significant.

Only 14 genes, including *Ccl21a*, showed significantly higher expression in EGFP⁺ cTECs than in EGFP⁻ cTECs, whereas only one gene, the parathyroid hormone-encoding *Pth*, showed significantly lower expression in EGFP⁺ cTECs than in EGFP⁻ cTECs (Fig. 4C, Fig. S3D). All the 14 genes that showed significantly higher expression in EGFP⁺ cTECs than in EGFP⁻ cTECs represented genes that showed higher expression in mTECs than in cTECs, and their expression, including *Ccl21a* expression, was much higher in mTECs than in EGFP⁺ cTECs (Fig. S3D). Interestingly, EGFP⁺ cTECs distributed unequally within the thymic cortex and were detected more frequently in the perimedullary cortical region than in the central cortical region or the subcapsular region of the cortex (Fig. 4D, E, Fig. S1C). In contrast, Ly51⁺ total cTECs did not show such an unequal distribution in the thymic cortex (Fig. S1D, E). Thus, the intrathymic proximity to the medullary region contributes to the previous expression of *Ccl21a* transcripts in cTECs, suggesting that approximately two-thirds of postnatal cTECs originate from *Ccl21a*⁺ cells generated in the thymic medulla.

Collectively, these fate-mapping experiments demonstrate that a vast majority of mTECs, including the self-antigen-displaying mTEC subset, and approximately two-thirds of cTECs, particularly cTECs in the perimedullary cortical region, are derived from progenitor cells that previously transcribed *Ccl21a*.

***Ccl21a*-expressing mTECs arise early in central region of thymus primordium**

It is further interesting to note that EGFP⁺ cells within Foxn1⁺ mTECs were detected in the central region of the thymus primordium in *Ccl21a*-Cre x CAG-loxP-EGFP mice as early as E11 (Fig. 5A). The detection of EGFP⁺ cells occurred earlier in the ontogeny than the detection of *Ccl21a*^{tdTomato}⁺ mTECs by E13 (Fig. 2B) and CCL21-protein-expressing mTECs by E15 (Fig. 2D). These sequential detections suggested that low-level *Ccl21a* transcription initiated as early as E11 during thymus organogenesis was sufficient to produce functional Cre proteins to mediate the expression of CAG-promoter-dependent EGFP proteins, but failed to produce a detectable amount of tdTomato or CCL21 proteins. Subsequent elevation of *Ccl21a* transcription level resulted in the production of detectable amounts of tdTomato proteins followed by CCL21 proteins. Indeed, single-cell RNA-sequencing analysis of *Ccl21a* transcript counts demonstrated that *Ccl21a* expression levels within individual *Epcam*⁺ Foxn1⁺ TECs elevated during the embryogenesis from E12 to E14 (Fig. 5B). At E14, *Ccl21a*^{low} cells were detected at an equivalent frequency in β5t⁺ and β5t⁻ TEC populations (Fig. 5B). *Ccl21a* transcription in CCL21-expressing mTECs was further elevated from embryonic to postnatal development, as shown by single-cell RNA-sequencing analysis of *Ccl21a* transcript counts (Fig. 5B) and verified by flow cytometric analysis of *Ccl21a*-dependent tdTomato expression levels (Fig. 5C). Importantly, *Ccl21a*⁺ Foxn1⁺ mTECs detected during early thymus organogenesis were localized in the central region of the thymus primordium and were anatomically and transcriptomically distinguishable from *Ccl21a*⁺ *Gcm2*⁺ parathyroid epithelial cells (Fig. 2B, 5A, 5D). These results reveal that *Ccl21a*⁺ mTECs develop in the central region of the thymus primordium as early as E11 before the generation of detectably CCL21 protein-expressing mTECs in the embryonic thymus by E15.

CCL21-expressing mTECs are distinct from RANK-expressing mTECs

Along with the fate-mapping detection of the developmental potential in *Ccl21a*⁺ cells, the early and temporal detection of *Ccl21a*⁺ mTECs and CCL21-expressing mTECs during embryogenesis tempted us to speculate that *Ccl21a*⁺ mTECs, and even CCL21-expressing mTECs, might possess mTEC progenitor activity. Previous studies reported that embryonic thymus develops RANK-expressing mTEC-restricted progenitors, which have a developmental capability to give rise to self-antigen-displaying mTECs, including Aire⁺ mTECs^{26,27}, RANK-expressing mTEC progenitors arise in the thymus during embryogenesis downstream of mTEC stem cells, which originate from cTEC-trait-expressing bipotent progenitors^{5,26,29}. We next examined whether CCL21-

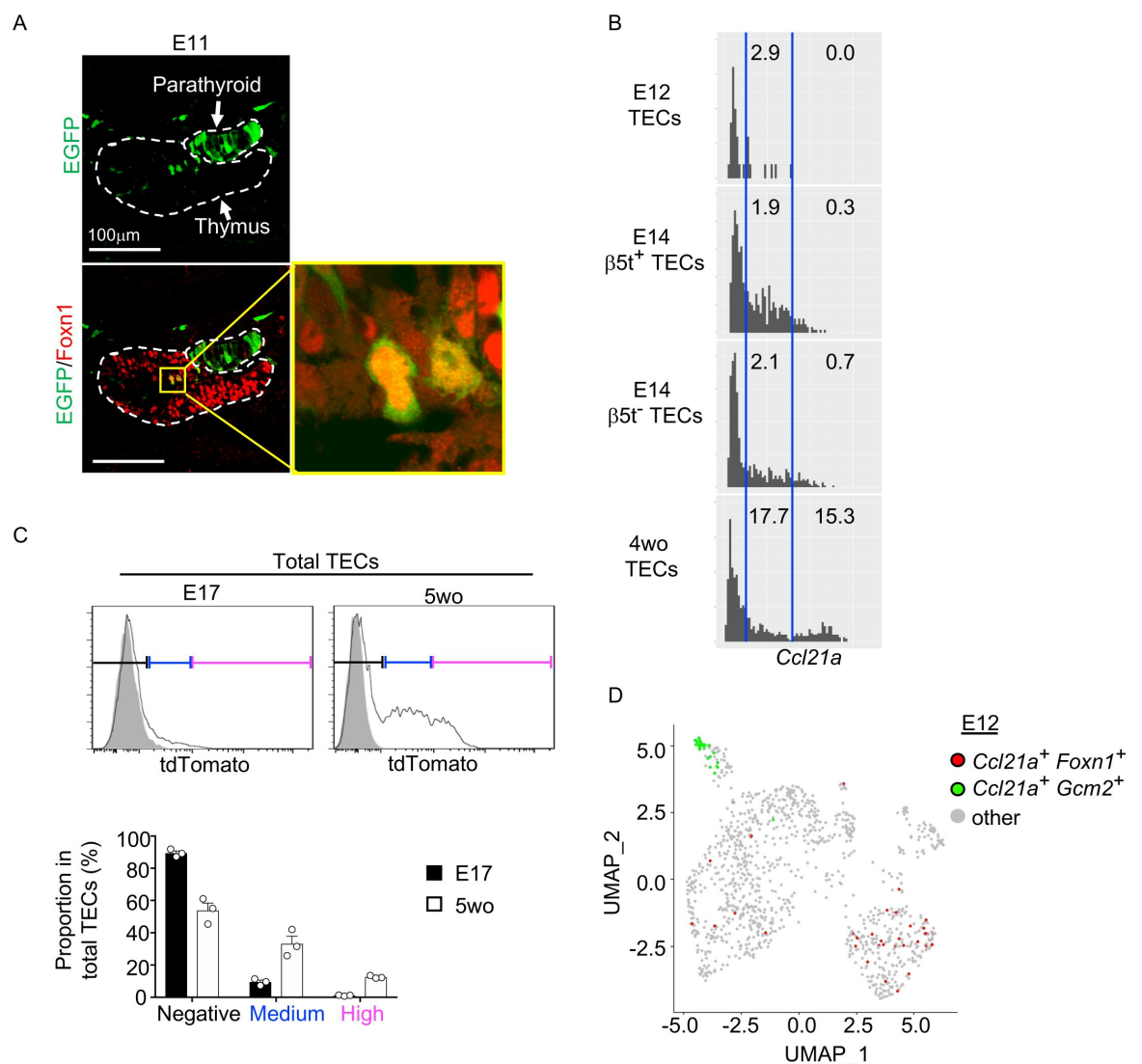


Figure 5.

Ccl21a-expressing mTECs during early thymus organogenesis

(A) Immunofluorescence analysis of thymus section from *Ccl21a*-Cre x CAG-loxP-EGFP E11 embryos. EGFP (green) and Foxn1 (red) were analyzed as indicated. Dashed lines show the outline of the thymus and parathyroid primordia. Image in yellow box in the middle of the thymus is magnified in right panel. Scale bar, 100µm. Representative data from two independent experiments are shown.

(B) Single-cell RNA-sequencing analysis of *Epcam*⁺*Foxn1*⁺ TECs from E12, E14, and 4-week-old $\beta 5t$ -Venus knock-in mice. Normalized log transcript counts of *Ccl21a* mRNA (x-axis) and cell numbers (y-axis) are plotted. For E14 TECs, $\beta 5t$ -Venus⁺ and $\beta 5t$ -Venus⁻ TECs were sorted and analyzed in parallel. Numbers in histograms indicate frequency of cells within indicated area.

(C) Flow cytometric analysis of tdTomato expression in *Epcam*⁺CD45⁻ TECs isolated from E17 embryonic and 2-week-old postnatal *Ccl21a*^{tdTomato/+} mice (black histograms) and WT mice (gray shaded histograms). Bottom plots indicate the proportions (means and SEMs, n=3) of tdTomato^{negative} (black), tdTomato^{medium} (blue), and tdTomato^{high} (magenta) cells within *Epcam*⁺CD45⁻ TECs as defined in top histograms.

(D) Single cell RNA-sequencing analysis of *Epcam*⁺ E12 embryonic pharyngeal epithelial cells. Dots indicate UMAP plot of *Ccl21a*⁺ *Foxn1*⁺ mTECs (red), *Ccl21a*⁺ *Gcm2*⁺ parathyroid epithelial cells (green), and other cells (gray).

expressing mTECs might overlap with RANK-expressing mTEC progenitors in *CCL21^{tdTomato} RANK^{Venus}* embryonic thymus. Interestingly, however, we found that CCL21-expressing mTECs and RANK-expressing mTECs were largely distinct cell populations within the embryonic thymus (**Fig. 6A, B**), although they were closely localized within the medullary region (**Fig. 6A**). RNA-sequencing analysis of CCL21⁺RANK⁻ and CCL21⁻RANK⁺ TECs highly purified from *CCL21^{tdTomato} RANK^{Venus}* embryonic thymus (Fig. S4A) demonstrated that CCL21-expressing mTECs were markedly different from RANK-expressing mTECs in global gene expression profiles (**Fig. 6C**). Gene ontology enrichment analysis ascertained the difference in global gene expression profiles in CCL21-expressing mTECs and RANK-expressing mTECs (**Fig. 6D**). Indeed, CCL21-expressing mTECs retained more cTEC-trait genes, including *Psm11*, *Cxcl12*, *CD83*, and *Ackr4*, than RANK-expressing mTECs (**Fig. 6E**), suggesting that in comparison with RANK-expressing mTECs, CCL21-expressing mTECs are more recently derived from cTEC-trait-expressing bipotent TEC progenitors. However, similarly to RANK-expressing mTECs, CCL21-expressing mTECs expressed equivalent levels of some mTEC-trait transcripts, including *Krt5* and *Krt14* (**Fig. 6E**). Among the genes associated with TEC progenitor activity^{30–35}, *Trp63* and *Pax1* were predominant in CCL21-expressing mTECs, whereas *Plet1*, *Krt19*, *Cldn3*, *Cldn4*, and *Ly6a* were readily detectable in RANK-expressing mTECs (**Fig. 6E**). Thus, CCL21-expressing mTECs in embryonic thymus are largely distinct from RANK-expressing mTECs, whereas CCL21-expressing mTECs and RANK-expressing mTECs are proximally localized with each other within the thymus medulla.

Nonetheless, it is important to point out that fate-mapping experiments using *Ccl21a*-Cre x CAG-loxP-EGFP mice were unable to distinguish whether TEC progenitor activity resides in CCL21-expressing mTECs generated within the thymus or in CCL21-expressing non-mTECs including neighboring parathyroid cells.

Developmental potential of CCL21-expressing mTECs

To directly examine whether CCL21-expressing mTECs have a developmental potential to give rise to self-antigen-displaying mTECs, we finally traced the fate of CCL21-expressing mTECs within the thymus microenvironment. To do so, we employed reaggregate thymus organ culture experiments^{36,37} where RelB-deficient thymic stromal cells do not generate any functional mTECs^{38,39}. Reaggregate thymuses were transplanted under mouse kidney capsules for reconstitution with hematopoietic cells derived from recipient mice. In this experimental condition, RelB-deficient thymic stromal cells were reaggregated with tdTomato-expressing mTECs highly purified from *Ccl21a^{tdTomato}* E17 embryos (Fig. S4B). *Ccl21a^{tdTomato}*⁺ mTECs had already become functional CCL21-protein-expressing mTECs by E17 (**Fig. 2**), and the accumulation of CD4⁺CD8⁻ and CD4⁺CD8⁺ thymocytes in E17 thymic medullary region was defective in mice deficient in CCL21 (**Fig. 7A–C**), so that the reaggregate thymus experiments would address whether functionally relevant CCL21-expressing mTECs give rise to other mTEC subpopulations including Aire⁺ mTECs.

We found that *Ccl21a^{tdTomato}*-positive EpCAM⁺CD45⁻ mTECs isolated from *Ccl21a^{tdTomato}*⁺ E17 embryos generated Aire⁺ Krt14⁺ mTECs in the thymus reaggregated with RelB-deficient thymic stromal cells (**Fig. 8A**). Aire⁺ Krt14⁺ mTECs also developed from an equal number of parallelly isolated *Ccl21a^{tdTomato}*-negative EpCAM⁺CD45⁻ TECs (**Fig. 8B**), which included cTEC-mTEC bipotent TEC progenitors^{5,23}. RelB-deficient thymic stromal reaggregates without additional TECs failed to generate Aire⁺ Krt14⁺ mTECs (**Fig. 8C**), whereas control organ culture using RelB-heterozygous thymus apparently developed medullary regions that contained Aire⁺ Krt14⁺ mTECs (**Fig. 8D**). Quantitative evaluation of Krt14⁺ medullary areas (**Fig. 8E**) within total thymus aggregates (**Fig. 8F**) demonstrated that *Ccl21a⁺* mTECs have a developmental potential to give rise to Aire⁺ Krt14⁺ mTECs, equivalently to and not statistically different from *Ccl21a*-negative TECs, including bipotent TEC progenitors (**Fig. 8G, H**). Interestingly, *Ccl21a⁺* mTECs were still detected in the Krt14⁺ medullary areas in reconstituted thymus reaggregated with either *Ccl21a⁺* mTECs or *Ccl21a*-negative TECs (**Fig. 8I, J**), indicating that *Ccl21a⁺* mTECs have a potential to maintain themselves in the thymus and that *Ccl21a*-negative TECs have a developmental potential

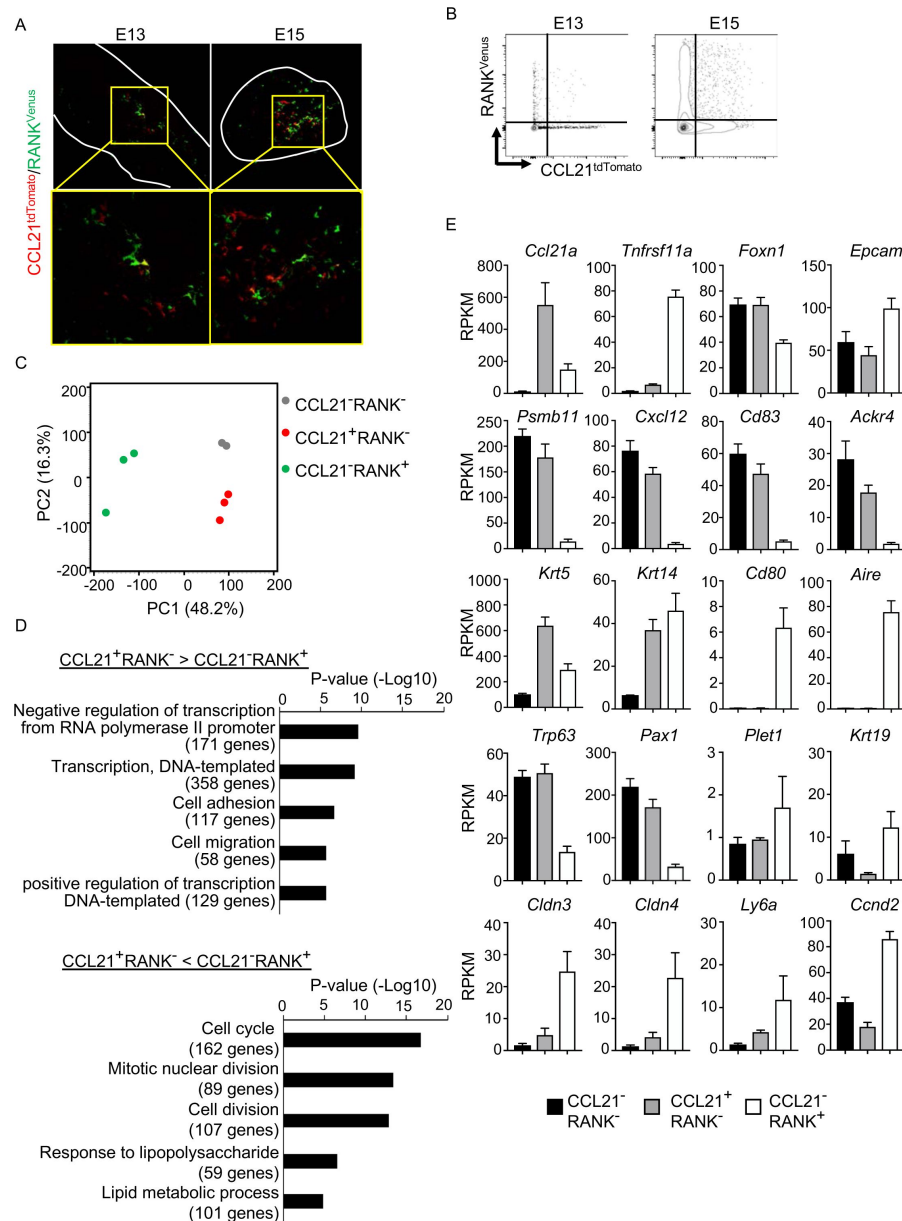


Figure 6.

CCL21-expressing mTECs are distinct from RANK-expressing mTECs

(A) Immunofluorescence analysis of tdTomato (red) and Venus (green) signals in thymus sections from *Ccl21^{tdTomato/+} RANK^{Venus}* mice at indicated embryonic age. White lines indicate the capsular outline of the thymus. Images in boxed regions in upper panels are magnified in bottom panels. Representative data from at least three independent experiments are shown.

(B) Flow cytometric profiles of tdTomato and Venus fluorescence signals in EpCAM⁺CD45⁻ TECs isolated from *Ccl21^{tdTomato/+} RANK^{Venus}* mice at indicated embryonic age.

(C) Principal component analysis of RNA-sequencing data of indicated cell populations isolated from *Ccl21^{tdTomato/+} RANK^{Venus}* E17 embryos.

(D) Enrichment analysis of ontology for genes that are differently expressed (RPKM > 1, fold change > 1.5 or < 1.5, FDR p-value < 0.05) between CCL21⁺RANK⁻ and CCL21⁺RANK⁺ TECs. Bars show the adjusted p-values of the top five categories enriched in CCL21⁺RANK⁻ TECs (top) and CCL21⁺RANK⁺ TECs (bottom). Numbers in parentheses indicate the number of categorized genes.

(E) RPKM values of indicated genes detected in RNA-sequencing analysis.

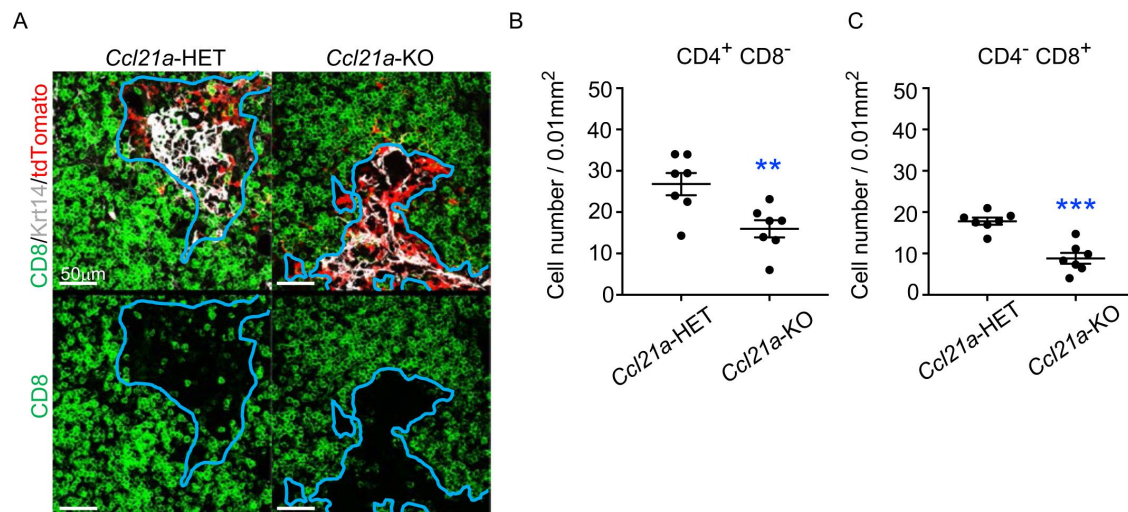


Figure 7.

CCL21-expressing mTECs in E17 embryos are functional in thymocyte attraction

(A) Immunofluorescence analysis of CD4, CD8, Krt14, and tdTomato in thymus sections from heterozygous (*Ccl21a*^{tdTomato/+}) control mice or homozygous (*Ccl21a*^{tdTomato/tdTomato}) CCL21-deficient mice at E17. E17 thymus sections were stained for CD4, CD8, and Krt14. Representative images for Krt14 (white), tdTomato fluorescence (red), and CD8 (green) expressions are shown. Blue lines indicate cortico-medullary junctions, according to the distribution of Krt14-expressing mTECs. tdTomato-positive cells were localized in the medullary regions. Scale bar, 50µm.

(B, C) Graphs show the means and SEMs (n = 3) of the numbers of CD4⁺CD8⁻ **(B)** and CD4⁻CD8⁺ **(C)** thymocytes per unit area (0.01 mm²) in the medullary regions. **, P < 0.01; ***, P < 0.001.

to produce *Ccl21a*⁺ mTECs. Most importantly, these results directly indicate that functionally relevant CCL21-expressing mTECs have a developmental potential to give rise to Aire-expressing mTECs.

In contrast to CCL21-expressing mTECs isolated from E17 embryos, *Ccl21a*^{tdTomato}-positive EpCAM⁺CD45[−] mTECs isolated from 4-week-old *Ccl21a*^{tdTomato/+} mice did not exhibit any signs of Aire⁺ Krt14⁺ mTEC generation in the thymus reaggregated with RelB-deficient thymic stromal cells (Fig. S5A–G), suggesting a decline in the developmental potential of CCL21⁺ mTECs during the ontogeny from embryonic to postnatal period. However, no detection of progeny could also result from technical caveats associated with reaggregate thymus organ culture conditions that were suitable for detection of the developmental potential of embryonic but not postnatal CCL21-expressing mTECs, or differences in the lifespan and/or turnover of embryonic and postnatal CCL21-expressing cells.

Throughout the development including E17 embryos, CCL21-expressing mTECs were localized specifically in the central region of the thymus and were Krt14⁺ (Fig. 1, 2, 6, 9A). However, single-cell RNA-sequencing analysis demonstrated that the global gene expression profiles were largely different between embryonic *Ccl21a*^{low} mTECs and 4-week-old postnatal *Ccl21a*^{high} mTECs (Fig. 9B). Indeed, the majority of CCL21-expressing mTECs at E17 but not the postnatal period simultaneously expressed cTEC-associated protein Ly51 and mTEC-associated UEA1-binding molecules on the cell surface (Fig. 9C). Ly51 expression was lower in embryonic TECs than postnatal TECs⁴⁰ and equivalently distributed in the thymus including the medullary region defined by Krt14-expressing mTECs centrally localized in E17 thymus (Fig. 9A). These results highlight the difference in molecular expression profiles, including Ly51 expression, between E17 CCL21-expressing mTECs and postnatal CCL21-expressing mTECs.

Discussion

The present results demonstrate that CCL21-expressing mTECs arise early during embryonic thymus development. *Ccl21a*-*Cre*-mediated detection reveals that *Ccl21a*⁺ *Foxn1*⁺ mTECs are detectable as early as E11 in the central region of the embryonic thymus, shortly after the organogenesis of the thymus primordium, and this is followed by the detection of *Ccl21a*-*tdTomato*-expressing mTECs by E13 and CCL21 protein-expressing mTECs by E15. *Ccl21a*-*Cre*-mediated fate-mapping analysis establishes that at 3 weeks old, a vast majority of mTECs, including the self-antigen-displaying mTECs such as Aire⁺ mTECs and thymic tuft cells, and approximately two-thirds of cTECs originate from *Ccl21a*⁺ cells. Most importantly, our results demonstrate that *Ccl21a*-*tdTomato*-expressing and functional CCL21 protein-producing mTECs isolated from E17 embryonic thymus have a developmental potential to give rise to the self-antigen-displaying mTECs in the thymus microenvironment. These results demonstrate a previously unknown process by which the diversity in the thymus medulla epithelium is generated during embryogenesis by the developmental conversion of the CCL21-expressing thymocyte-attracting mTEC subset into the self-antigen-displaying mTEC subset, including Aire-expressing mTECs and thymic tuft cells.

Functional conversion of the thymocyte-attracting mTEC subset into the self-antigen-displaying mTEC subset offers an interesting implication that the thymus medulla is formed by initially generating CCL21-expressing mTECs and attracting CCR7-expressing positively selected thymocytes from the cortex even before the subsequent display of self-antigens mediated by Aire⁺ mTECs and thymic mimetic cells. Failure to establish self-tolerance by the loss of *Ccr7* or *Ccl21a* in the thymus^{10,12} underscored the importance of functional CCL21-expressing mTEC subset. Our present results showing developmental potential of functional CCL21-expressing mTECs to give rise to self-antigen-displaying mTECs reveals an additionally important function of CCL21-expressing mTECs in forming the diversity in thymus medulla epithelium, including the self-

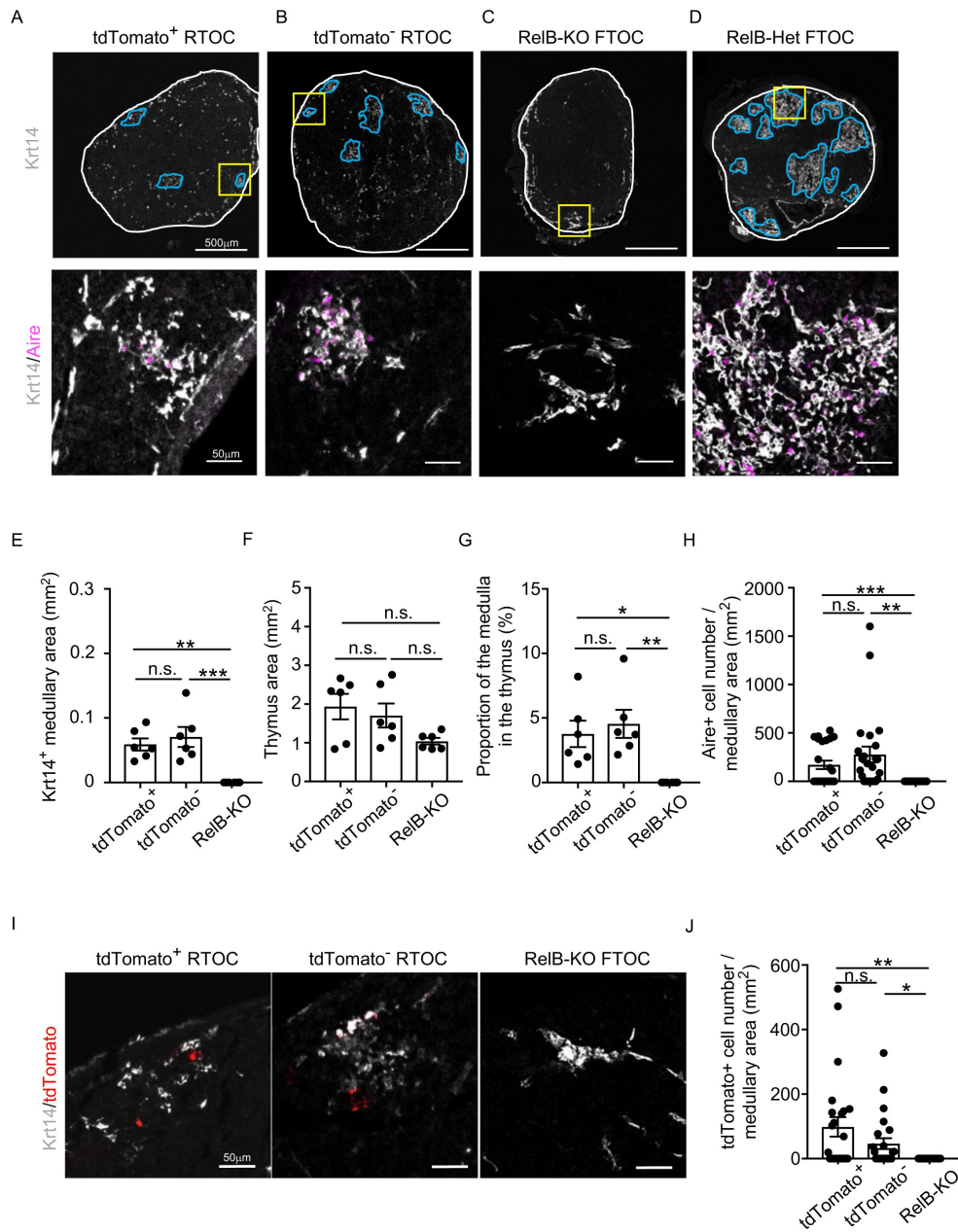


Figure 8.

Developmental potential of CCL21-expressing mTECs

(A-D) Immunofluorescent staining of indicated thymus graft sections. Shown in upper panels are Krt14 expression (white). White lines indicate capsular outline of the thymus. Blue lines show Krt14⁺ medullary region. Scale bar, 500μm. Bottom panels show Krt14 (white) and Aire (magenta) fluorescence signals within yellow boxes in upper panels. Scale bar, 50μm. Grafts represent RelB-KO thymic stroma reaggregated with tdTomato⁺ TECs (A) and tdTomato⁻ TECs (B) as well as fetal thymus lobe from RelB-KO mice (C) and RelB-heterozygous mice (D).

(E) Size of Krt14⁺ medullary areas (means and SEMs, n=6) in indicated thymus graft sections.

(F) Size of grafted thymus areas (means and SEMs, n=6) in indicated thymus graft sections.

(G) Proportion (means and SEMs, n=6) of Krt14⁺ medullary areas in the thymus areas in indicated thymus graft sections.

(H) Numbers of Aire⁺ mTECs per mm² of Krt14⁺ medullary areas in indicated thymus graft sections.

(I) Immunofluorescence analysis of tdTomato (red) and Krt14 (white) in indicated thymus graft sections. Scale bar, 50μm.

(J) Numbers of tdTomato⁺ cells per mm² of Krt14⁺ medullary areas in indicated thymus graft sections. All images are representative data from three independent experiments. *P<0.05, **P<0.01, ***P<0.001, n.s., not significant.

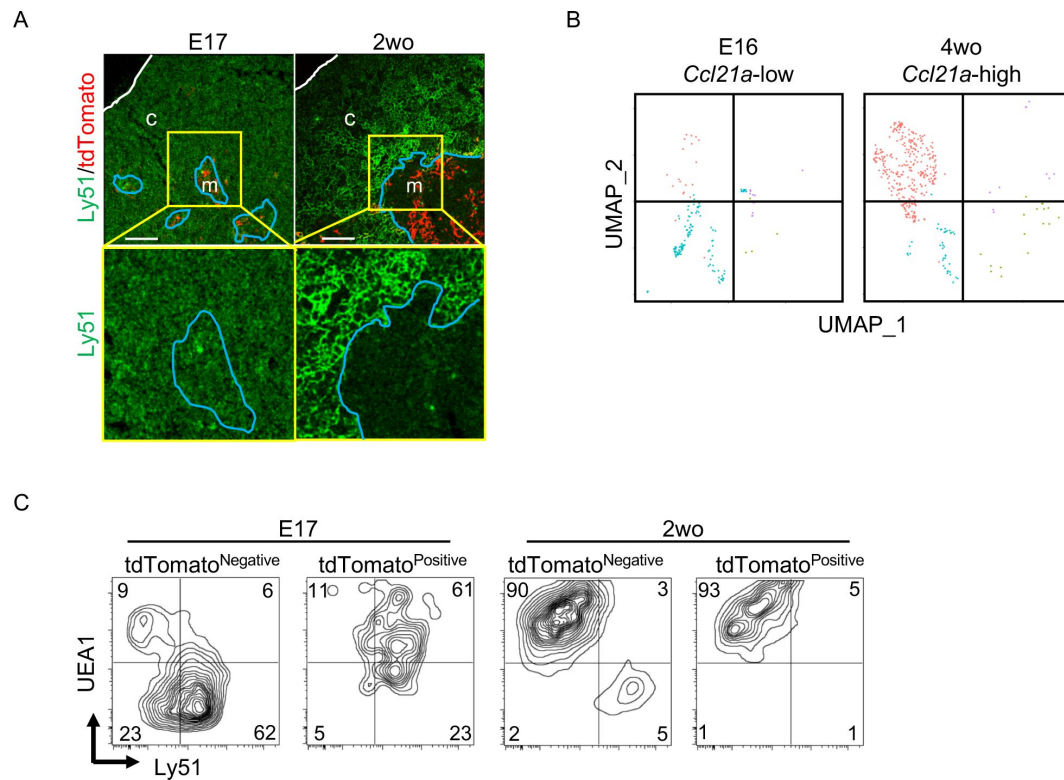


Figure 9.

CCL21-expressing mTECs at E17 embryonic and postnatal period

(A) Immunofluorescence analysis of thymus sections from E17 or 2-week-old *Ccl21a*^{tdTomato/+} mice. tdTomato fluorescence (red) and Ly51 (green) expression are shown. White lines indicate thymic capsules. Blue lines indicate cortico-medullary junctions. Images in yellow boxes (top panels) are magnified in bottom panels. Representative data from at least three independent experiments are shown. c, cortex. m, medulla. Scale bar, 100μm.

(B) UMAP plots showing the clusters of *Ccl21a*⁺ mTECs in single cell RNA-sequencing analysis of thymic epithelial cells isolated from E16 embryonic and 4-week-old postnatal mice. The majority of E16 *Ccl21a*⁺ mTECs were *Ccl21a*^{low}, whereas the majority of postnatal *Ccl21a*⁺ mTECs were *Ccl21a*^{high} (Fig. 5B, C [↗](#)).

(C) Flow cytometric profiles of UEA1 reactivity and Ly51 expression in tdTomato^{negative} EpCAM⁺CD45⁻ TECs and tdTomato^{positive} EpCAM⁺CD45⁻ TECs isolated from E17 embryonic and 2-week-old postnatal *Ccl21a*^{tdTomato/+} mice. Numbers in the contour plots indicate frequency of cells within indicated area. Representative data from three independent experiments are shown.

antigen-displaying mTEC subset. Stepwise progression in mTEC development from thymocyte-attracting function to self-antigen-displaying function fits stepwise requirement to initially recruit newly generated T cells from the thymic cortex before tolerizing T cells to self-antigens. Down-regulation of CCL21 in functionally converted self-antigen-displaying mTECs may further contribute to the prevention of excessive negative selection of newly generated T cells to self-antigens.

Recent studies demonstrated that the function of the thymus medulla is not limited to the establishment of self-tolerance in conventional $\alpha\beta$ T cells but extends to the development of innate lymphocytes, including invariant NKT cells and $\gamma\delta$ T cells^{41,42}. The developmental capability of CCL21-expressing mTECs may further extend to additional mTEC subpopulations that contribute to supporting the development of innate invariant lymphocytes. Thus, CCL21-expressing mTECs may represent a primordial mTECs which separate newly generated $\alpha\beta$ T cells, $\gamma\delta$ T cells and NKT cells from the thymic cortex to the medullary microenvironment for further selection and maturation.

In contrast to the robust generation of Aire⁺ mTECs from embryonic CCL21⁺ mTECs, our results show that the ability to give rise to Aire⁺ mTECs was not detectable in postnatal CCL21⁺ mTECs. The fact that no Aire⁺ mTECs were detected in reaggregate thymus organ culture may reflect a postnatal decline in the developmental potential of CCL21⁺ mTECs. However, it is also possible that the developmental kinetics and the requirements for postnatal CCL21⁺ mTECs may be different from those for E17 CCL21⁺ mTECs, so that the reaggregate thymus organ culture conditions for detecting the development of E17 CCL21⁺ mTECs may not be optimal for the detection of the survival and development of postnatal CCL21⁺ mTECs.

Nevertheless, it is important to notice the difference in the developmental potential between embryonic and postnatal CCL21⁺ mTECs in reaggregate thymus organ culture. Analysis of molecular expression profiles further highlighted the difference between embryonic and postnatal CCL21⁺ mTECs. We think that CCL21⁺ mTECs may be heterogenous in developmental capability. Embryonic CCL21⁺ mTECs, which give rise to self-antigen-displaying mTECs including Aire⁺ mTECs and mimetic mTECs, may also terminally differentiate into a subpopulation of CCL21⁺ mTECs, which no longer have further developmental potential, and which represent the majority of postnatal CCL21⁺ mTECs.

Interestingly, our results also reveal that approximately two-thirds of cTECs are derived from progenitor cells that previously transcribed *Ccl21a*. The previous expression of *Ccl21a* transcripts in the fraction of cTECs is influenced by the intrathymic proximity to the medullary region and is enriched in the perimedullary cortical region (Fig. 4E). Developmental capability to become cTECs is shown in *Ccl21a* gene-expressing cells, and it is unknown whether CCL21-protein-expressing functional mTECs retain the developmental potential to give rise to cTECs. In contrast, our results from reaggregate thymus experiments clarified that CCL21-protein-expressing functional mTECs have a developmental potential to convert into Aire-expressing mTECs. Consequently, we think that at least a fraction of *Ccl21a* transcript-expressing mTECs retain a bipotent developmental potential to give rise to both cTECs and mTECs. It is unclear whether most *Ccl21a*-expressing mTECs retain a bipotent developmental potential for cTECs and mTECs or *Ccl21a*-expressing mTECs represent a mixture of bipotent progenitors and lineage-committed mTEC progenitors. Because of the unequal distribution of EGFP-labeled cTECs within the thymic cortex, initial signals to promote bipotent progenitors to mTEC lineage may be provided by molecules restricted in the central region of the embryonic thymus, and those signals may promote the transcription of *Ccl21a* at an early stage of mTECs, which still retain the potential to become cTECs. Further examination of cTEC potential, for example, by using CCL21-tdTomato-expressing cells that constitutively express GFP, may lead to a better understanding of the cTEC-generating capability in embryonic and postnatal CCL21⁺ mTECs. It is also interesting to examine

whether GFP⁺ cTEC development in *Ccl21a*-Cre x CAG-loxP-EGFP mice is mediated through RelB-dependent mTEC developmental progression and/or dependent on developing thymocyte-dependent mTEC-nurturing ‘crosstalk’ signals.

Nusser, et al., recently identified two bipotent TEC progenitor populations: an early cTEC-biased progenitor population and a postnatal mTEC-biased progenitor population²³. Interestingly, they noted that the postnatally appearing mTEC-biased progenitor population includes cells that transcribe *Ccl21a*. It would be interesting to clarify whether and how those postnatal TEC progenitors overlap with the embryonic and postnatal CCL21-protein-expressing mTECs reported in this study. It would also be interesting to shed light on how differently *Ccl21a*⁺ progenitors contribute to cTECs and mTECs over the ontogeny and whether the enrichment of *Ccl21a*⁺ progenitor-derived cTECs in the perimedullary area reflects a temporal replacement of cTECs derived from *Ccl21a*⁺ progenitors localized in the medulla.

Further characterization of CCL21-expressing mTECs carrying a developmental potential to give rise to self-antigen-displaying mTEC subset is of great interest. Our results highlight the difference between CCL21-expressing mTECs and previously characterized RANK-expressing mTEC-restricted progenitors. CCL21-expressing mTECs and RANK-expressing mTECs are detectable in E13 embryonic thymus, are proximally localized within the medullary region in the thymus, and are capable of generating Aire-expressing mTECs. Unlike RANK-expressing mTECs, however, CCL21-expressing mTECs include developmental potential to give rise to cTECs. Gene expression profiles show that CCL21-expressing mTECs retained more cTEC-trait genes, including *Psb11*, *Cxcl12*, *CD83*, and *Akr4*, than RANK-expressing mTECs, suggesting that in comparison with RANK-expressing mTECs, CCL21-expressing mTECs are more closely related to and more recently derived from cTEC-trait-expressing bipotent TEC progenitors. Furthermore, it is interesting to note that recently described Krt19⁺ progenitors that give rise to multiple mature mTEC populations including Aire⁺ mTECs, mimetic mTECs, and postnatal CCL21⁺ mTECs, lack expression of CCL21 and RANK³³. An important aspect of future work will be to examine whether CCL21-expressing mTECs, RANK-expressing mTECs and Krt19-expressing mTECs are linked in upstream-downstream relationships in a single developmental pathway, or are developmentally distinct and parallel progenitors to give rise to mTECs. Similarly, it will be interesting to better clarify possible heterogeneity in CCL21-expressing mTECs in relation to their ability to give rise to cTECs as well as mTECs. Nevertheless, the finding in our study that CCL21⁺ embryonic mTECs can give rise to Aire⁺ mTECs reveals a new source of an important component of the thymus medulla.

Our results show that 95% of mTECs are labeled with EGFP fluorescence and therefore derived from *Ccl21a*⁺ progenitors, whereas 5% of mTECs are not labeled with EGFP fluorescence in *Ccl21a*-Cre x loxP-EGFP mice. Our previous study using β5t-driven Cre mice indicated fate-mapping of 95% EGFP⁺ mTECs in CAG-loxP-stop-loxP-EGFP-transgenic reporter mice used in this study and >99.5% ZsGreen⁺ mTECs in Rosa26-loxP-ZsGreen-knock-in reporter mice⁴³. Consequently, we think that 95% EGFP⁺ cells in mTECs detected in this study may indicate that virtually all (>99.5%) mTECs are derived from cells that once transcribed *Ccl21a*.

A previous study detected *Ccl21a* transcripts within a podoplanin-expressing TEC subpopulation that resembled immature TECs in postnatal thymus⁴⁴. It was reported that podoplanin-expressing mTEC progenitors, also known as junctional TECs (jTECs), were localized at the corticomedullary junction⁴⁴. However, our results show that CCL21-expressing mTECs in the embryonic and postnatal thymus distribute throughout the medullary region and are not enriched in the corticomedullary junctions (**Fig. 1H**, **2G**). Indeed, the majority of CCL21-expressing mTECs in postnatal thymus did not express podoplanin (Fig. S5H). Moreover, podoplanin expression was broadly detected by the majority of embryonic E17 TECs, most of which were cTECs (Fig. S5H). Thus, neither podoplanin expression nor corticomedullary junctional distribution characterizes CCL21-expressing mTECs. More importantly, our results demonstrate

developmental potential in thymocyte-attracting CCL21-expressing mTECs and identify developmental conversion between functionally distinct mTEC subsets, which serves to assemble a diverse medullary microenvironment in the thymus.

In conclusion, the present study establishes that CCL21-expressing functional mTEC subset carries a developmental potential to give rise to self-antigen-displaying mTECs, revealing that the sequential conversion of functionally distinct mTEC subsets from thymocyte-attracting cells into self-antigen-displaying cells contributes to the functional diversification of thymus medulla epithelium during embryonic development. The present results are likely to ignite many interesting questions, including the characterization of signals that drive the developmental conversion of mTECs and the elucidation of developmental pathways for diverse TEC subpopulations.

Materials and methods

Mice

C57BL/6 (B6) mice were obtained from The Jackson Laboratory (Bar Harbor, ME) and SLC (Shizuoka, Japan). *Ccl21a*-tdTomato knock-in mice¹⁰, RelB-knockout mice³⁹, RANK-Venus transgenic mice⁴⁵, β 5t-Venus knock-in mice⁴⁶, CAG-loxP-stop-loxP-EGFP transgenic mice⁴⁷, and Foxn1-GFP transgenic mice⁴⁸ were described previously. All mouse experiments were performed with consent from the Animal Experimentation Committee of the University of Tokushima (T2022–50), the Animal Care and Use Committee of the National Cancer Institute (ASP 21–431, ASP 21–432, and EIB-076–3), the Birmingham Animal Welfare and Ethical Review Board and UK Home Office Animal Licences (PP7518148 and PP2990911), and the Institutional Animal Care and Use Committee of RIKEN Kobe Branch (A2001-03).

Generation of *Ccl21a*-Cre mice

The targeting vector was prepared by subcloning *Ccl21a*-containing mouse genomic BAC fragments (Advanced Genotechs) and Cre-encoding cDNA into a plasmid containing a pgk-neo cassette (Gene Bridges). The linearized targeting vector was introduced into TT2 ES cells⁴⁹. Targeted knock-in alleles were screened by genomic PCR analysis and Southern blot analysis. The *Ccl21a*-Cre mice (<https://large.riken.jp/distribution/mutant-list.html>; Accession No. CDB1131K) are available to the scientific community and were backcrossed to B6 mouse strain for more than 5 generations. The knock-in allele is detected using PCR primers as follows; Cre-F, 5'-AGGTTCGTTCACTCATGGA-3' and Cre-R, 5'-TCGACCAGTTTAGTTACCC-3' (product size, 235bp). Wild-type allele is detected using PCR primers as follows; WT-F, 5'-CTGGTCTCATCCTCAACTCA-3' and WT-R, 5'-TGTAACCCTAGGATTGTAGG-3' (product size, 1,563bp).

Reaggregate thymus experiments

Fetal thymus lobes isolated from RelB-KO mice at embryonic day (E) 15 were organ cultured for 5 days on sponge-supported Nucleopore filters (Whatman) placed on Dulbecco's Modified Eagle Medium supplemented with 10% fetal bovine serum (FBS) and penicillin-streptomycin mixed solution. To eliminate endogenous hematopoietic cells from the lobes, 1.35 mM 2'-deoxyguanosine (dGuo) was added in the culture medium^{36,37}. dGuo-treated thymus lobes were digested with 0.25% trypsin supplemented with 0.45 mM EDTA and 0.02% DNase I for 15 min at 37°C and reaggregated with tdTomato⁺ or tdTomato[−] TECs isolated from *Ccl21a*^{tdTomato/+} mice at E17 or 4 weeks old. After one day culture on Nucleopore filters, reaggregated thymuses were transplanted under kidney capsules of B6 mice. dGuo-treated fetal thymus lobes isolated from RelB-deficient and RelB-heterozygous mice were also transplanted under kidney capsules of B6 mice. Grafts were harvested 5 weeks post transplantation.

Multicolor immunofluorescence analysis

Paraformaldehyde-fixed frozen tissues embedded in OCT compound (Sakura Finetek) were sliced into 10- μ m-thick sections. The sections were stained with anti-CCL21 (Bio-Rad), anti-Aire (eBioscience), anti-Ly51 (Biolegend), anti-Krt14 (Biolegend), anti-DCLK1 (abcam), anti-CD4 (Biolegend), anti-CD8 (Biolegend) antibodies, anti-Foxn1 antiserum⁵⁰, and UEA-1 (Vector Laboratories). Images were analyzed with a TCS SP8 confocal laser scanning microscope (Leica).

Fluorescence distribution analysis

The plot profile intensity for the fluorescence image of each channel in the regions of interest (ROI) was measured using ImageJ software (<https://imagej.nih.gov/ij/index.html>). The fluorescence intensity at each horizontal distance with the total intensity of the vertically averaged tdTomato, EGFP, or Ly51 in the ROI was calculated as 100. The sum of tdTomato or EGFP intensities for the medullary region defined by Krt14 expression or UEA1 reactivity and for the cortical region was calculated.

Multicolor flow cytometric analysis and cell sorting

For the analysis of thymic epithelial cells (TECs), minced thymuses were digested with 0.5 unit/ml Liberase (Roche) in the presence of 0.02% DNase I (Roche). Single-cell suspensions were stained with antibodies specific for EpCAM (BioLegend), CD45 (eBioscience), Ly51 (Biolegend), Podoplanin (Biolegend), I-A^b (BioLegend), and for the reactivity with UEA-1 (Vector Laboratories). For the analysis of Aire and DCLK1, surface-stained cells were fixed in 5% formaldehyde neutral buffer solution (Nacalai Tesque), permeabilized in 1x permeabilization buffer (eBioscience), and stained with anti-Aire (eBioscience) or anti-DCLK1 (abcam) antibody. For the isolation of TECs, CD45⁺ cells were enriched in magnetic bead conjugated anti-CD45 antibody (Miltenyi Biotec). Multicolor flow cytometry and cell sorting were performed on FACSVerse, FACSARIA II, and LSRFortessa (BD Biosciences).

RNA-sequencing analysis

cDNAs were prepared by using SMART-Seq v4 Ultra Low Input RNA Kit, according to the manufacturer's protocol (Clontech). Sequencing libraries were generated by using a Nextera XT DNA Library Prep Kit, according to the manufacturer's protocol (Illumina). The concentration of libraries was measured by an ABI PRISM 7500 Real-time PCR system in combination with a Power SYBR Green PCR Master Mix (Thermo Fisher). Single-end sequencing of cDNA libraries with a read length of 51 was performed with HiSeq 1500 platform (Illumina). For RNA-sequencing analysis of CCL21⁺RANK⁺ and CCL21⁺RANK⁺ TECs isolated from CCL21^{tdTomato} RANK^{Venus} embryonic E17 thymus, RNA was prepared using Qiagen RNeasy Plus Micro Kit with genomic DNA eliminator columns. RNA-sequencing samples were sequenced on NovaSeq_SP flowcell using Illumina TruSeq Stranded Total RNA Library Prep with 2 $\times$ 100bp paired-end sequencing. Data were analyzed by using CLC Genomics Workbench 12.0 (Qiagen) with default parameters.

Single cell RNA-sequencing analysis

E12 embryonic pharyngeal cells, E14 embryonic CD45⁺EpCAM⁺ β 5t-Venus⁺ and β 5t-Venus⁺ TECs, E16 embryonic CD45⁺EpCAM⁺ β 5t-Venus⁺ and β 5t-Venus⁺ TECs isolated from β 5t-Venus knock-in mice, and 4-week-old postnatal TEC-enriched thymus cells were employed for single-cell RNA sequencing. 3' mRNA-seq library was generated as outlined in the Chromium Connect platform using 3' v3.1 Single Cell RNA-Seq chemistry reagents from 10X Genomics. Reverse transcription and cDNA amplification were performed according to the manufacturer's instructions. Libraries were sequenced on NextSeq 2000 and FASTQ files were generated from demultiplexed raw base call files using Cell Ranger 6.1.1 mkfastq. Alignment, filtering, barcode counting, and UMI counting were performed using Cell Ranger 6.1.1 count with mouse reference 3.0.0 (mm10 and Ensembl release 98). The resulting matrix was imported into Seurat (v4.3.0) in R v 4.3.0 to create a Seurat

object. The object underwent normalization, filtering, scaling, and dimensionality reduction steps sequentially using Seurat build-in functions, with default parameters. To integrate all the filtered objects from each sample, the CCA algorithm was employed through the FindIntegrationAnchors and IntegrateData functions in Seurat. The Uniform Manifold Approximation and Projection (UMAP) plots were visualized using the DimPlot function of Seurat. The histograms were visualized using the ggplot2 package (v3.4.2).

Statistical analysis

Statistical significance was assessed using the two-tailed unpaired Student's t test with Welch's correction for unequal variances or one-way ANOVA with Tukey's correction.

Data availability

RNA sequencing data are deposited in The DNA Data Bank of Japan (<https://www.ddbj.nig.ac.jp>) BioProject database with the accession number PRJDB15439 and NCBI Gene Expression Omnibus (GEO) (<https://www.ncbi.nlm.nih.gov/geo>) database with the accession number GSE239913. Single-cell RNA-sequencing data are deposited in NCBI GEO with the accession number GSE243180.

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References

1. Derbinski J, Schulte A, Kyewski B, Klein L (2001) **Promiscuous gene expression in medullary thymic epithelial cells mirrors the peripheral self** *Nat Immunol* **2**:1032–9 <https://doi.org/10.1038/ni723>
2. Anderson MS *et al.* (2002) **Projection of an immunological self shadow within the thymus by the aire protein** *Science* **298**:1395–401 <https://doi.org/10.1126/science.1075958>
3. Farr AG, Dooley JL, Erickson M (2002) **Organization of thymic medullary epithelial heterogeneity: implications for mechanisms of epithelial differentiation** *Immunol Rev* **189**:20–7 <https://doi.org/10.1034/j.1600-065x.2002.18903.x>
4. Michelson DA, Hase K, Kaisho T, Benoist C, Mathis D (2022) **Thymic epithelial cells co-opt lineage-defining transcription factors to eliminate autoreactive T cells** *Cell* **185**:2542–2558 <https://doi.org/10.1016/j.cell.2022.05.018>
5. Takahama Y, Ohigashi I, Baik S, Anderson G (2017) **Generation of diversity in thymic epithelial cells** *Nat Rev Immunol* **17**:295–305 <https://doi.org/10.1038/nri.2017.12>
6. Ueno T, Saito F, Gray DH, Kuse S, Hieshima K, Nakano H, Kakiuchi T, Lipp M, Boyd RL, Takahama Y (2004) **CCR7 signals are essential for cortex-medulla migration of developing thymocytes** *J Exp Med* **200**:493–505 <https://doi.org/10.1084/jem.20040643>
7. Takahama Y (2006) **Journey through the thymus: stromal guides for T-cell development and selection** *Nat Rev Immunol* **6**:127–35 <https://doi.org/10.1038/nri1781>
8. Nakano H, Gunn MD (2001) **Gene duplications at the chemokine locus on mouse chromosome 4: multiple strain-specific haplotypes and the deletion of secondary lymphoid-organ chemokine and EBI-1 ligand chemokine genes in the plt mutation** *J Immunol* **166**:361–9 <https://doi.org/10.4049/jimmunol.166.1.361>
9. Chen SC, Vassileva G, Kinsley D, Holzmann S, Manfra D, Wiekowski MT, Romani N, Lira SA (2002) **Ectopic expression of the murine chemokines CCL21a and CCL21b induces the formation of lymph node-like structures in pancreas, but not skin, of transgenic mice** *J Immunol* **168**:1001–8 <https://doi.org/10.4049/jimmunol.168.3.1001>
10. Kozai M, Kubo Y, Katakai T, Kondo H, Kiyonari H, Schaeuble K, Luther SA, Ishimaru N, Ohigashi I, Takahama Y (2017) **Essential role of CCL21 in establishment of central self-tolerance in T cells** *J Exp Med* **214**:1925–1935 <https://doi.org/10.1084/jem.20161864>
11. Kurobe H *et al.* (2006) **CCR7-dependent cortex-to-medulla migration of positively selected thymocytes is essential for establishing central tolerance** *Immunity* **24**:165–77 <https://doi.org/10.1016/j.immuni.2005.12.011>
12. Nitta T, Nitta S, Lei Y, Lipp M, Takahama Y (2009) **CCR7-mediated migration of developing thymocytes to the medulla is essential for negative selection to tissue-restricted antigens** *Proc Natl Acad Sci U S A* **106**:17129–33 <https://doi.org/10.1073/pnas.0906956106>

13. Link A, Vogt TK, Favre S, Britschgi MR, Acha-Orbea H, Hinz B, Cyster JG, Luther SA (2007) **Fibroblastic reticular cells in lymph nodes regulate the homeostasis of naive T cells** *Nat Immunol* **8**:1255–65 <https://doi.org/10.1038/ni1513>
14. Derbinski J, Gäbler J, Brors B, Tierling S, Jonnakuty S, Hergenhausen M, Peltonen L, Walter J, Kyewski B (2005) **Promiscuous gene expression in thymic epithelial cells is regulated at multiple levels** *J Exp Med* **202**:33–45 <https://doi.org/10.1084/jem.20050471>
15. Gray DH, Seach N, Ueno T, Milton MK, Liston A, Lew AM, Goodnow CC, Boyd RL (2006) **Developmental kinetics, turnover, and stimulatory capacity of thymic epithelial cells** *Blood* **108**:3777–85 <https://doi.org/10.1182/blood-2006-02-004531>
16. Miller CN *et al.* (2018) **Thymic tuft cells promote an IL-4-enriched medulla and shape thymocyte development** *Nature* **559**:627–631 <https://doi.org/10.1038/s41586-018-0345-2>
17. Kadouri N, Nevo S, Goldfarb Y, Abramson J (2020) **Thymic epithelial cell heterogeneity: TEC by TEC** *Nat Rev Immunol* **20**:239–253 <https://doi.org/10.1038/s41577-019-0238-0>
18. Metzger TC, Khan IS, Gardner JM, Mouchess ML, Johannes KP, Krawisz AK, Skrzypczynska KM, Anderson MS (2013) **Lineage tracing and cell ablation identify a post-Aire-expressing thymic epithelial cell population** *Cell Rep* **5**:166–79 <https://doi.org/10.1016/j.celrep.2013.08.038>
19. Rossi SW *et al.* (2007) **RANK signals from CD4+3– inducer cells regulate development of Aire-expressing epithelial cells in the thymic medulla** *J Exp Med* **204**:1267–72 <https://doi.org/10.1084/jem.20062497>
20. Gäbler J, Arnold J, Kyewski B (2007) **Promiscuous gene expression and the developmental dynamics of medullary thymic epithelial cells** *Eur J Immunol* **37**:3363–72 <https://doi.org/10.1002/eji.200737131>
21. Lkhagvasuren E, Sakata M, Ohigashi I, Takahama Y (2013) **Lymphotoxin β receptor regulates the development of CCL21-expressing subset of postnatal medullary thymic epithelial cells** *J Immunol* **190**:5110–7 <https://doi.org/10.4049/jimmunol.1203203>
22. Baran-Gale J *et al.* (2020) **Ageing compromises mouse thymus function and remodels epithelial cell differentiation** *Elife* **9** <https://doi.org/10.7554/eLife.56221>
23. Nusser A Sagar, Swann JB, Krauth B, Diekhoff D, Calderon L, Happe C, Grün D, Boehm T (2022) **Developmental dynamics of two bipotent thymic epithelial progenitor types** *Nature* **606**:165–171 <https://doi.org/10.1038/s41586-022-04752-8>
24. Wells KL, Miller CN, Gschwind AR, Wei W, Phipps JD, Anderson MS, Steinmetz LM (2020) **Combined transient ablation and single-cell RNA-sequencing reveals the development of medullary thymic epithelial cells** *Elife* **9** <https://doi.org/10.7554/eLife.60188>
25. Liu C, Saito F, Liu Z, Lei Y, Uehara S, Love P, Lipp M, Kondo S, Manley N, Takahama Y (2006) **Coordination between CCR7- and CCR9-mediated chemokine signals in prevascular fetal thymus colonization** *Blood* **108**:2531–9 <https://doi.org/10.1182/blood-2006-05-024190>
26. Baik S, Sekai M, Hamazaki Y, Jenkinson WE, Anderson G (2016) **Relb acts downstream of medullary thymic epithelial stem cells and is essential for the emergence of RANK+ medullary epithelial progenitors** *Eur J Immunol* **46**:857–62 <https://doi.org/10.1002/eji.201546253>

27. Akiyama N *et al.* (2016) **Identification of embryonic precursor cells that differentiate into thymic epithelial cells expressing autoimmune regulator** *J Exp Med* **213**:1441–58 <https://doi.org/10.1084/jem.20151780>
28. Sekai M, Hamazaki Y, Minato N (2014) **Medullary thymic epithelial stem cells maintain a functional thymus to ensure lifelong central T cell tolerance** *Immunity* **41**:753–61 <https://doi.org/10.1016/j.immuni.2014.10.011>
29. Ohigashi I, Zuklys S, Sakata M, Mayer CE, Hamazaki Y, Minato N, Hollander GA, Takahama Y (2015) **Adult thymic medullary epithelium is maintained and regenerated by lineage-restricted cells rather than bipotent progenitors** *Cell Rep* **13**:1432–1443 <https://doi.org/10.1016/j.celrep.2015.10.012>
30. Senoo M, Pinto F, Crum CP, McKeon F (2007) **p63 Is essential for the proliferative potential of stem cells in stratified epithelia** *Cell* **129**:523–36 <https://doi.org/10.1016/j.cell.2007.02.045>
31. Wallin J, Eibel H, Neubüser A, Wilting J, Koseki H, Balling R (1996) **Pax1 is expressed during development of the thymus epithelium and is required for normal T-cell maturation** *Development* **122**:23–30 <https://doi.org/10.1242/dev.122.1.23>
32. Ulyanchenko S, O'Neill KE, Medley T, Farley AM, Vaidya HJ, Cook AM, Blair NF, Blackburn CC (2016) **Identification of a bipotent epithelial progenitor population in the adult thymus** *Cell Rep* **14**:2819–32 <https://doi.org/10.1016/j.celrep.2016.02.080>
33. Lucas B *et al.* (2023) **Embryonic keratin19+ progenitors generate multiple functionally distinct progeny to maintain epithelial diversity in the adult thymus medulla** *Nat Commun* **14** <https://doi.org/10.1038/s41467-023-37589-4>
34. Hamazaki Y, Fujita H, Kobayashi T, Choi Y, Scott HS, Matsumoto M, Minato N (2007) **Medullary thymic epithelial cells expressing Aire represent a unique lineage derived from cells expressing claudin** *Nat Immunol* **8**:304–11 <https://doi.org/10.1038/ni1438>
35. Wong K, Lister NL, Barsanti M, Lim JMC, Hammett MV, Khong DK, Siatskas C, Gray DHD, Boyd RL, Chidgey AP (2014) **Multilineage potential and self-renewal define an epithelial progenitor cell population in the adult thymus** *Cell Rep* **8**:1198–209 <https://doi.org/10.1016/j.celrep.2014.07.029>
36. Anderson G, Jenkinson EJ, Moore NC, Owen JJ (1993) **MHC class II-positive epithelium and mesenchyme cells are both required for T-cell development in the thymus** *Nature* **362**:70–3 <https://doi.org/10.1038/362070a0>
37. White A, Jenkinson E, Anderson G. (2008) **Reaggregate thymus cultures** *J Vis Exp* <https://doi.org/10.3791/905>
38. Weih F, Carrasco D, Durham SK, Barton DS, Rizzo CA, Ryseck RP, Lira SA, Bravo R (1995) **Multiorgan inflammation and hematopoietic abnormalities in mice with a targeted disruption of RelB, a member of the NF-kappa B/Rel family** *Cell* **80**:331–40 [https://doi.org/10.1016/0092-8674\(95\)90416-6](https://doi.org/10.1016/0092-8674(95)90416-6)
39. Burkly L, Hession C, Ogata L, Reilly C, Marconi LA, Olson D, Tizard R, Cate R, Lo D (1995) **Expression of relB is required for the development of thymic medulla and dendritic cells** *Nature* **373**:531–6 <https://doi.org/10.1038/373531a0>

40. Ripen AM, Nitta T, Murata S, Tanaka K, Takahama Y (2011) **Ontogeny of thymic cortical epithelial cells expressing the thymoproteasome subunit $\beta 5t$** *Eur J Immunol* **41**:1278–87 <https://doi.org/10.1002/eji.201041375>
41. White AJ, Jenkinson WE, Cowan JE, Parnell SM, Bacon A, Jones ND, Jenkinson EJ, Anderson G (2014) **An essential role for medullary thymic epithelial cells during the intrathymic development of invariant NKT cells** *J Immunol* **192**:2659–66 <https://doi.org/10.4049/jimmunol.1303057>
42. Roberts NA *et al.* (2012) **Rank signaling links the development of invariant $\gamma\delta$ T cell progenitors and Aire⁺ medullary epithelium** *Immunity* **36**:427–37 <https://doi.org/10.1016/j.immuni.2012.01.016>
43. Ohigashi I, Zuklys S, Sakata M, Mayer CE, Zhanybekova S, Murata S, Tanaka K, Holländer GA, Takahama Y (2013) **Aire-expressing thymic medullary epithelial cells originate from $\beta 5t$ -expressing progenitor cells** *Proc Natl Acad Sci U S A* **110**:9885–90 <https://doi.org/10.1073/pnas.1301799110>
44. Onder L *et al.* (2015) **Alternative NF- κ B signaling regulates mTEC differentiation from podoplanin-expressing precursors in the cortico-medullary junction** *Eur J Immunol* **45**:2218–31 <https://doi.org/10.1002/eji.201545677>
45. McCarthy NI, Cowan JE, Nakamura K, Bacon A, Baik S, White AJ, Parnell SM, Jenkinson EJ, Jenkinson WE, Anderson G (2015) **Osteoprotegerin-mediated homeostasis of Rank⁺ thymic epithelial cells does not limit Foxp3⁺ regulatory T cell development** *J Immunol* **195**:2675–82 <https://doi.org/10.4049/jimmunol.1501226>
46. Murata S, Sasaki K, Kishimoto T, Niwa S, Hayashi H, Takahama Y, Tanaka K (2007) **Regulation of CD8⁺ T cell development by thymus-specific proteasomes** *Science* **316**:1337–53 <https://doi.org/10.1126/science.1141915>
47. Kawamoto S, Niwa H, Tashiro F, Sano S, Kondoh G, Takeda J, Tabayashi K, Miyazaki J (2000) **A novel reporter mouse strain that expresses enhanced green fluorescent protein upon Cre-mediated recombination** *FEBS Lett* **470**:263–8 [https://doi.org/10.1016/S0014-5793\(00\)01338-7](https://doi.org/10.1016/S0014-5793(00)01338-7)
48. Terszowski G, Müller SM, Bleul CC, Blum C, Schirmbeck R, Reimann J, Pasquier LD, Amagai T, Boehm T, Rodewald HR (2006) **Evidence for a functional second thymus in mice** *Science* **312**:284–7 <https://doi.org/10.1126/science.1123497>
49. Yagi T, Tokunaga T, Furuta Y, Nada S, Yoshida M, Tsukada T, Saga Y, Takeda N, Ikawa Y, Aizawa S (1993) **A novel ES cell line, TT2, with high germline-differentiating potency** *Anal Biochem* **214**:70–76 <https://doi.org/10.1006/abio.1993.1458>
50. Itoi M, Tsukamoto N, Amagai T (2007) **Expression of Dll4 and CCL25 in Foxn1-negative epithelial cells in the post-natal thymus** *Int Immunol* **19**:127–32 <https://doi.org/10.1093/intimm/dxl129>

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Reviewer #1 (Public Review):

The work by Ohigashi and colleagues addresses the developmental and lineage relationship of a newly characterized thymus epithelial cell (TEC) progenitor subset. The authors take advantage of an elegant and powerful set of experimental approaches to demonstrate that CCL21-expressing TECs appear early in thymus organogenesis and that these cells, which are centrally located, go on to give rise to medullary (m)TECs. What makes the findings intriguing is that these CCL21-expressing mTECs are a distinct subset, which do not express RANK or AIRE, and transcriptomic and lineage tracing approaches point to these cells as potential mTEC progenitor-like cells. Of note, using in vitro and in vivo precursor-product cell transfer experiments, the authors show that this subset has a developmental potential to give rise to AIRE⁺ self-antigen-displaying mTECs, revealing that CCL21-expressing mTECs can give rise to distinct mTEC subsets. This functional duality provides an attractive rationale for the necessary function of mTECs, which is to attract CCR7⁺ thymocytes that have just undergone positive selection in the thymus cortex to enter the medulla to undergo tolerance-induction against self-antigen-displaying mTECs. Overall, the work is well supported and offers new insights into the diverse functions of the medullary compartment, and how two distinct subsets of mTECs can achieve it.

<https://doi.org/10.7554/eLife.92552.2.sa2>

Reviewer #2 (Public Review):

The authors set out to discover a developmental pathway leading to functionally diverse mTEC subsets. They show that Ccl21 is expressed early during thymus ontogeny in the medullary area. Fate-mapping gives evidence for the Ccl21 positive history of Aire positive mTECs as well as of thymic tuft cells and postnatally of a certain percentage of cTECs. Therefore, the differentiation potential of Ccl21⁺ TECs is tested in reaggregate thymus experiments - using embryonic or postnatal Ccl21⁺ TECs. From these experiments, the

authors conclude that at least embryonic mTECs in large part pass through a Ccl21 positive stage prior to differentiation towards an Aire expressing or tuft cell stage.

The authors are using Ccl21a as a marker for a bipotent progenitor that is detectable in the embryonic thymus and is still present at the adult stage mainly giving rise to mTECs. The choice of this marker gene is very interesting since Ccl21 expression can directly be linked to an important aspect in thymus biology: the expression of Ccl21 by cells in the thymic medulla allows trafficking of T cells into the medulla in order to undergo T cell selection. Making use of the Ccl21 detection, the authors can nicely show that cells actively expressing Ccl21 are localized throughout the medulla at an embryonic stage but also in adult thymus tissue. This suggests, that this progenitor is not accumulating at a specific area inside the medulla. This is a new finding. Moreover, the finding that a Ccl21+ progenitor population plays a functional role in thymocyte trafficking towards the medulla has not been described. Thus, Ccl21 expression may be used to localize a late bipotent progenitor in the thymic lobes. In addition, in Fig.8, the authors provide evidence that these progenitor cells have the potential to self-maintain as well as to differentiate in reaggregate experiments at E17 (not at 4 weeks of age). The first point is of great interest and importance since these cells in theory can be of therapeutic use.

Overall assessment:

The authors highlight a developmental pathway starting from a Ccl21-expressing TEC progenitor that contributes to a functionally diverse mTEC repertoire. This is a welcome addition to current knowledge of TEC differentiation.

<https://doi.org/10.7554/eLife.92552.2.sa1>

Reviewer #3 (Public Review):

In this manuscript, the authors define the developmental trajectory resulting in a diverse mTEC compartment. Using a variety of approaches, including a novel CCL21-fate mapping model, data is presented to argue that embryonic CCL21-expressing thymocyte attracting mTECs naturally convert to into self-antigen displaying mTEC subsets, including Aire+ mTECs and thymic tuft cells. Perhaps somewhat surprisingly, a large fraction of cTECs were also marked for having expressed CCL21, suggesting that there exists some conversion of mTEC (progenitors) into cTEC, a developmentally interesting observation that could be followed up later. Overall, the experimental setup, writing, and conclusions, are all outstanding. The one question I have, which may be more of a curiosity of this reviewer than a requirement for the manuscript, is whether thymocytes themselves are required for the conversion/maturation of attracting TECs to mTECs? For example, in CD3e-/- (or Rag-/-) mice, are mTECs arrested at the thymocyte attracting stage, or is the conversion process 'pre-programed'? In the same vein, do cTECs (or the immature cTECs) maintain CCL21 expression in the absence of mature thymocytes? These are not critical studies but are fairly straightforward (effort- and time-wise) that would aid in placing this process in the overall scope of thymus development.

<https://doi.org/10.7554/eLife.92552.2.sa0>

Author Response

The following is the authors' response to the original reviews.

eLife assessment

This is an important study that provides new insights into the development and function of medullary thymus epithelial cells (mTEC). The authors provide compelling evidence to support their claims as to the differentiation and lineage outcomes of CCL21+ mTEC progenitors, which further our understanding of how central tolerance of T cells is enforced within the thymus.

Public Reviews:

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Reviewer #2 (Public Review):

Summary:

The authors set out to discover a developmental pathway leading to functionally diverse mTEC subsets. They show that Ccl21 is expressed early during thymus ontogeny in the medullary area. Fate-mapping gives evidence for the Ccl21 positive history of Aire positive mTECs as well as of thymic tuft cells and postnatally of a certain percentage of cTECs. Therefore, the differentiation potential of Ccl21+ TECs is tested in reaggregate thymus experiments - using embryonic or postnatal Ccl21+ TECs. From these experiments, the authors conclude that at least embryonic mTECs in large part pass through a Ccl21 positive stage prior to differentiation towards an Aire expressing or tuft cell stage.

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Moreover, the finding that a Ccl21⁺ progenitor population plays a functional role in thymocyte trafficking towards the medulla has not been described. Thus, Ccl21 expression may be used to localize a late bipotent progenitor in the thymic lobes.

In addition, in Fig.8, the authors provide evidence that these progenitor cells have the potential to self-maintain as well as to differentiate in reaggregate experiments at E17 (not at 4 weeks of age). The first point is of great interest and importance since these cells in theory can be of therapeutic use.

Overall assessment:

The authors highlight a developmental pathway starting from a Ccl21-expressing TEC progenitor that contributes to a functionally diverse mTEC repertoire. This is a welcome addition to current knowledge of TEC differentiation.

Reviewer #3 (Public Review):

In this manuscript, the authors define the developmental trajectory resulting in a diverse mTEC compartment. Using a variety of approaches, including a novel CCL21-fate mapping model, data is presented to argue that embryonic CCL21-expressing thymocyte attracting mTECs naturally convert to into self-antigen displaying mTEC subsets, including Aire⁺ mTECs and thymic tuft cells. Perhaps somewhat surprisingly, a large fraction of cTECs were also marked for having expressed CCL21, suggesting that there exists some conversion of mTEC (progenitors) into cTEC, a developmentally interesting observation that could be followed up later. Overall, the experimental setup, writing, and conclusions, are all outstanding.

Provisional author response

We thank the editors and the reviewers for their supportive comments on our manuscript. We will revise the manuscript according to their helpful recommendations.

Author response to recommendations

We thank the editors and the reviewers for their supportive comments on our manuscript. We also thank the three reviewers for their helpful recommendations. We have revised the manuscript accordingly, as detailed below.

Reviewer #1 (Recommendations For The Authors):

There are several unanswered questions, which the authors themselves acknowledge, a principal one being whether CCL21⁺ mTECs represent a progenitor for yet another distinct subset of cortical (c)TECs, or whether they represent an intermediary or unique population of mTECs derived from a bipotent (cTEC/mTEC) progenitor. These questions will need to be addressed in future work as they go beyond the initial characterization of this intriguing mTEC subset.

Indeed, our findings reported in this manuscript have stimulated many interesting questions, including those pointed out by the reviewer. We would like to address them one by one in our future work.

The presence of GFP⁺ cTECs, which are lineage-traced as having expressed CCL21, begs the question as to whether these cells are generated as a consequence of later steps in mTEC differentiation or derived from earlier bipotent cells, which again the authors point out. The authors could discuss this further or perhaps experimentally address this by

using a model system whereby mTEC differentiation is absent or halted (e.g., Relb ko, or TCRa/TCRd ko) and test whether GFP+ cTECs are still present.

According to the suggestion, we have revised the manuscript by adding a statement that it is interesting to examine whether GFP+ cTEC development in Ccl21a-Cre x CAG-loxP-EGFP mice is mediated through RelB-dependent mTEC developmental progression or developing thymocyte-dependent mTEC-nurturing 'crosstalk' signals.

Reviewer #2 (Recommendations For The Authors):

Even though the manuscript highlights the functional aspect of a postnatal bipotent progenitor, there are several aspects that need further discussion.

(1) The title is somewhat misleading since the identified TEC subset can not only be detected in embryonic, but also in postnatal thymus. Only the RTOC experiments indicate a higher developmental potential of TECs isolated from embryos, but this might as well be due to experimental difficulties as discussed in the text. Furthermore, Ccl21+ TECs are shown to differentiate postnatally into mTECs and cTECs, therefore this subset presumably belongs to a bipotent progenitor population described earlier (their ref. 22, 39).

We are fully aware of previous studies showing that mTEC progenitors include cells that transcribe Ccl21a, and have cited them in the manuscript. The manuscript title describes our finding that thymocyte-attracting CCL21-expressing functional mTECs isolated from embryonic thymus show the capability to give rise to self-antigen-displaying mTECs. We thank the reviewer for further pointing out the possibility that postnatal CCL21+ TECs include cells that retain the capability to differentiate into mTECs and cTECs.

(2) In the introduction the authors claim that the "developmental progression of the self-antigen-displaying mTEC subset occurs in a single stream as mTEC_{low} progenitors -> mTEC_{high} Aire-expressing cells -> mTEC_{low} mimetic cells." line 79. So far it only could be shown that some mimetic cell types undergo an Aire+ stage; whether this is true for all mimetic cells remains to be shown. Therefore, this statement should be toned down.

Following the suggestion, this sentence has been toned down in the revised manuscript.

(3) In line 86, the reference to another paper, describing Ccl21a expression in a postnatal mTEC biased progenitor should be added: Nusser et al. Nature. 2022 PMID: 35614226, in which the developmental potential of the Ccl21 positive so-called postnatal progenitor is analysed by barcoding and results give evidence for differentiation into mature mTECs (see lines 94-96).

As suggested, the Introduction of the revised manuscript now cites Nusser, et al. study showing that postnatal mTEC-biased progenitors include cells that transcribe Ccl21a.

(4) Have a look at Extended Data Figure 2b of PMID: 35614226, wherein the population-specific gene expression pattern of the progenitor population at different time points is depicted. Ccl21a belongs to a group of genes, which identifies the postnatal progenitor, and indicates that its functionality and/or developmental potential is age-dependent. Therefore, it would be important to specify the age of the analysed mice throughout the text of the results part instead of describing them as "postnatal" only.

As recommended, mouse age has been added to the revised manuscript and figures.

(5) Line 113: "embryonic" needs to be replaced since the results of Fig. 1 are referring to 5-week-old mice.

The manuscript has been revised per the reviewer's suggestion.

(6) Referring to Fig. 3g, line 173: It is interesting to see that, at 3 weeks of age, 95% of mTECs have a Ccl21-history but only approx. 70% of cTECs. Therefore, the earliest progenitor giving rise to the first cTECs might still be productive and feed into the cTEC lineage. This reporter would allow for the analysis of progenitor activity over time. The same could be done for mTECs since at E15 the tdTomato signal is still low compared to the assigned medullary area in Fig. 2c in order to detect when the Ccl21-expressing progenitor becomes the main source of mTECs. The finding in Fig. 4e (line196) also argues for the timed replacement of cTECs by a progenitor which locates to the medulla, thus, leading to a decline in Ccl21-history signal towards the subcapsular region at 2 weeks of age. This should be better explained/discussed.

We appreciate the work of Nusser, et al. showing that postnatal mTEC-biased, but not embryonic cTEC-biased, TEC progenitors include cells that transcribe a detectable amount of Ccl21a (cited in the Introduction as ref. 23). It is important to clarify whether and how those postnatal TEC progenitors (23) overlap with the embryonic and postnatal CCL21-protein-expressing mTECs reported in this study. It is also interesting to shed light on how Ccl21a+ progenitors contribute to cTECs and mTECs over the ontogeny and whether the enrichment of Ccl21a+ progenitor-derived cTECs in the perimedullary area reflects a temporal replacement of cTECs derived from Ccl21a+ progenitors localized in the medulla. We would like to clarify these issues in our future work. The revised manuscript includes a discussion of these issues.

(7) Line 304 and 355: Note that the "unstable" age-dependent gene expression profiles were already reported in Nusser et al. Nature. 2022. Not only Ccl21 expression, but other progenitor-specific genes also change their expression levels with age. The entirety of changes in gene expression during aging likely impacts the developmental potential of progenitor populations. These changes might be reflected in the negative results of the RTOC experiment using TECs of 4-week-old mice. The manuscript would benefit from a discussion in light of this "unstable" age-dependent gene expression.

It is interesting to point out that the age-dependent difference in gene expression profiles, which was reported in TEC progenitors by Nusser, et al. (23), is also detected in CCL21-expressing mTECs in this study. Similarly to the recommendation no. 6 by reviewer 2, and as described in the revised manuscript, it is interesting to clarify whether and how embryonic and postnatal CCL21-expressing mTECs overlap with the previously reported TEC progenitors.

(8) Line 321: as discussed above, the exact time point should be added to the text since the proportion of cTECs derived from a Ccl21+ progenitor is associated with a certain time point, "2/3 of cTECs" refers to 3 weeks of age.

The manuscript has been revised following the reviewer's suggestion.

Reviewer #3 (Recommendations For The Authors):

The one question I have, which may be more of a curiosity of this reviewer than a requirement for the manuscript, is whether thymocytes themselves are required for the conversion/maturation of attracting TECs to mTECs? For example, in CD3e-/- (or Rag-/-)

mice, are mTECs arrested at the thymocyte attracting stage, or is the conversion process 'pre-programmed'? In the same vein, do cTECs (or the immature cTECs) maintain CCL21 expression in the absence of mature thymocytes? These are not critical studies but are fairly straightforward (effort- and time-wise) that would aid in placing this process in the overall scope of thymus development.

We previously showed that Aire⁺ mTECs are detectable in the thymus of RAG2-deficient mice, in which thymocyte development is arrested beyond the CD4/CD8 double-negative 3 stage (Hikosaka, et al. 2006; PMID: 18799150). In another work, we also showed that Aire⁺ mTECs and CCL21⁺ mTECs are detectable in the thymus of TCR-alpha-KO mice, which lack mature CD4/CD8 single-positive TCR-alpha/beta-expressing thymocytes (Lkhagvasuren, et al. 2013; PMID: 23585674). These results indicate that thymocyte maturation beyond the Rag-dependent stage is not essential for the development of Aire⁺ mTECs. Nonetheless, we agree with the reviewer pointing out that it is important to clarify how developing thymocytes contribute to the growth and differentiation of diverse TEC subpopulations, including GFP⁺ cTEC development in Ccl21a-Cre x CAG-loxP-EGFP mice. The revised manuscript includes a discussion of these issues.